



UNIVERSITY
OF TRENTO - Italy

Dipartimento di Ingegneria e Scienza dell'Informazione



LoE (T2MP)

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- **Theory recap**
- LoE Theories, Models, and Interpretation functions
- Types of problems
- I2SF and SF2I
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- LoE entailment

Theory Recap - Interpretation function

Definition (Interpretation function) Let L_a be a language of assertions and D a domain. Then an **Interpretation Function** I_a is defined as

$$I_a : L_a \rightarrow D$$

We say that a fact $f \in M$ is **the interpretation of** $a \in L_a$, and write

$$f = I_a(a) = a^I$$

to mean that the assertion a is a linguistic description of f .

We say that f is **the interpretation of** the assertion a , or, equivalently, that a **denotes** f .

Observation (Interpretation function). Interpretation functions apply to assertions and generate facts. Different world models refine the definition of interpretation function based on the percepts they consider. See later, the definition of the interpretation functions of the different world logics. This is a key element distinguishing one logic from another.

Theory Recap - Interpretation function

Observation (Non-ambiguity) Interpretation functions, being functions, are not ambiguous. If two facts f_1 and f_2 are different it cannot be that $I_a(a) = f_1$ and $I_a(a) = f_2$.

Observation (Interpretation function, synonymy). Two assertions are synonyms when they have the same meaning, that is, the interpretation of two different assertions a_1 and a_2 , may denote the same fact, i.e., $I_a(a_1) = I_a(a_2)$. In logic synonymy is not a problem as the denotation of a word is a single percept.

Observation (Totality). Interpretation functions are total. This guarantees that any element of the language has an interpretation.

Observation (Non-surjectivity). Interpretation functions are not necessarily surjective. In other words, if $I_a : L_a \rightarrow D$, L_a may not be able to name all the facts in D . This property formalizes the fact that linguistic representations do not necessarily describe all facts.

Theory Recap - Interpretation function

Definition (Language correctness and completeness). Let L_a and D be an assertional language and a domain of interpretation, respectively. Let $I_a : L_a \rightarrow D$ be an interpretation function. Then we have two possible situations, as follows

Language correctness. Let $a \in L_a$ be an assertion. If for all a , if $a \in L_a$ then $I_a(a) = f \in D$, then we say that L_a is **correct** with respect to D , or that D is a **domain** for L_a ;

Language completeness. Let $f \in D$ be a fact. If, for all f , if $f \in D$ then there is an assertion $a \in L_a$ such that $I_a(a) = f$, then we say that L_a is **complete** with respect to D .

Theory recap - World Logics

Proposition (World Logic). Let $W' = \langle L_{a'}, D', I_{a'} \rangle$ be a World Model. Then a **World Logic** L_W for W is defined as

$$L_W = \langle W, \models_W \rangle$$

with $W = \langle L_a, D, I_a \rangle$ and $L_{a'} \subseteq L_a$, $D' \subseteq D$ and $I_{a'} \subseteq I_a$.

Theory recap - LoE

Definition (LoE)

$$\text{LoE} = \langle \text{EG}, |=_{\text{LoE}} \rangle$$

with

$$\text{EG} = \langle L_{\text{LoE}}, D_{\text{LoE}}, I_{\text{LoE}} \rangle$$

When no confusion arises, we drop the subscripts.

Theory recap - LoE Domain

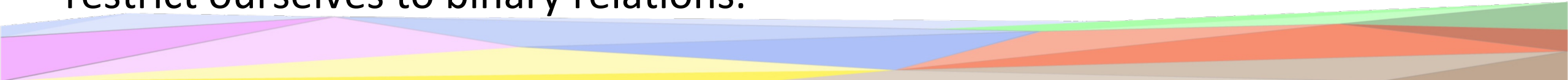
Definition (Domain)

$$D = \langle U, \{C\}, \{R\} \rangle$$

where:

- $U = \{u\}$ is called the **universe (of interpretation)** of D .
- $\{u\}$ is a set of **units** $u_1, \dots, u_n$, for some n
- $\{C\}$ is a set of **classes** $C_1, \dots, C_m$ of units, for some m , with $C_i \subseteq U$
- $\{R\}$ is a set of **binary relations** $R_1, \dots, R_p$ between units, for some p , with $R_i \subseteq U \times U$

Observation (EG, Binary relations). To comply with the graph notation we restrict ourselves to binary relations.



Theory Recap - LoE BNF

$\langle \text{assertion} \rangle$	$::= \langle \text{etype} \rangle (\langle \text{nameEntity} \rangle$	
	$\langle \text{dtype} \rangle (\langle \text{value} \rangle$	
	$\langle \text{objProp} \rangle (\langle \text{nameEntity} \rangle, \langle \text{nameEntity} \rangle)$	
	$\langle \text{dataProp} \rangle (\langle \text{nameEntity} \rangle, \langle \text{value} \rangle)$	
$\langle \text{etype} \rangle$	$::= E_1 \mid \dots \mid E_n$	
$\langle \text{dtype} \rangle$	$::= D_1 \mid \dots \mid D_n$	
$\langle \text{objProp} \rangle$	$::= O_1 \mid \dots \mid O_n$	
$\langle \text{dataProp} \rangle$	$::= A_1 \mid \dots \mid A_n$	
$\langle \text{nameEntity} \rangle$	$::= e_1 \mid \dots \mid e_n$	
$\langle \text{value} \rangle$	$::= v_1 \mid \dots \mid v_n$	

Theory Recap - LoE Interpretation function

Definition (Interpretation function, alphabet). The LoE interpretation function I is defined as

$$I : L \rightarrow D$$

where I is defined as

$$I = \langle I_U, I_T, I_P \rangle$$

where:

- $I_U = \langle I_e, I_v \rangle$, the universe interpretation function, is the pair of the entity and the value interpretation function;
- $I_T = \langle I_E, I_D \rangle$, the Type interpretation function, is the pair of the etype and the dtypes interpretation function;
- $I_P = \langle I_O, I_A \rangle$, the Property interpretation function, is the pair of the object property and the attribute interpretation function

with:

$$I_e : \{e\} \rightarrow \{e\}$$

$$I_E : \{E_i\} \rightarrow \{E_i\}$$

$$I_O : \{O_i\} \rightarrow \{O_i\}$$

$$I_v : \{v\} \rightarrow \{v\}$$

$$I_D : \{D_i\} \rightarrow \{D_i\}$$

$$I_A : \{A_i\} \rightarrow \{A_i\}$$

Theory Recap - LoE Interpretation function

Definition (LoE interpretation function, assertion). The interpretation of assertions into facts is defined as follows:

$$\begin{aligned} I(E_i(e)) &= I_E(E_i)(I_e(e)) &= e \in E_i \\ I(D_i(v)) &= I_D(D_i)(I_v(v)) &= v \in D_i \\ I(O_i(e_1, e_2)) &= I_O(O_i)(I_e(e_1), I_e(e_2)) &= \langle e_1, e_2 \rangle \in O_i \\ I(A_i(e, v)) &= I_A(A_i)(I_e(e), I_v(v)) &= \langle e, v \rangle \in A_i \end{aligned}$$

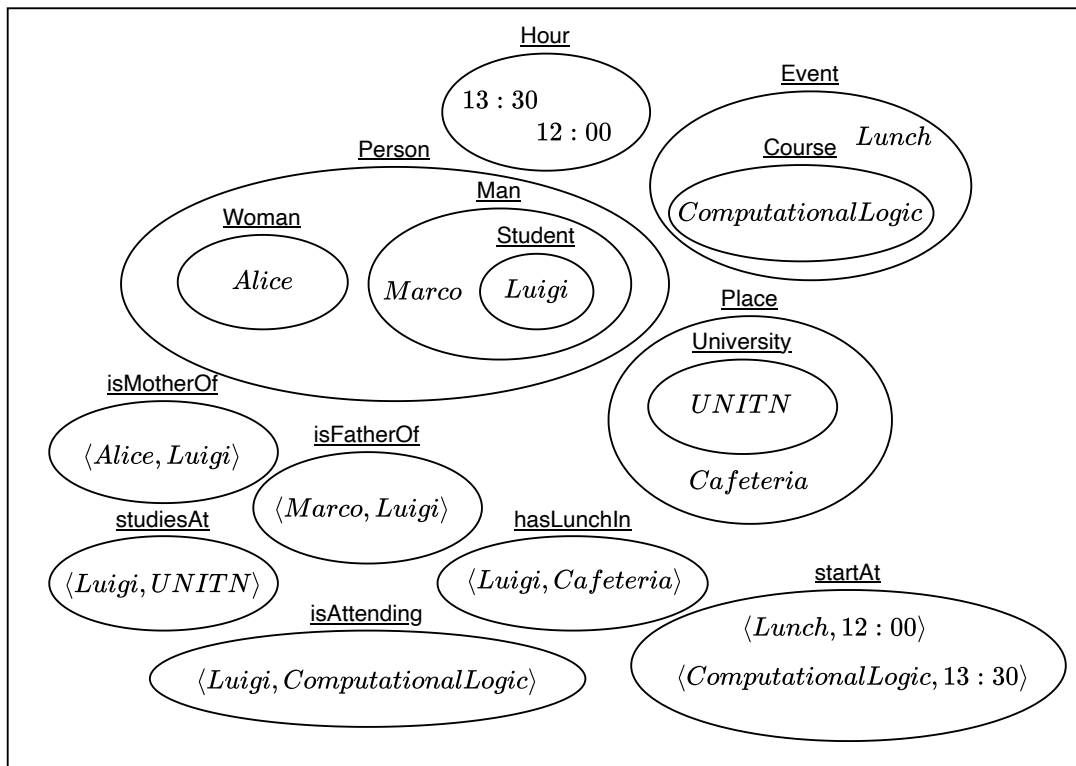
Observation. Notice how the interpretation of assertions first applies the interpretation function (i.e., I) which then, **compositionally**, applies the proper component specific interpretation function mimicking, step by step how syntax composes (e.g., in the first assertion, I_E and I_e).

Notation. The words of the alphabet are in “*italic*”, percepts are in “roman”

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Ex1 - Correctness of a Theory



Considering the intended model on the left and the interpretation function below, is the following a theory correct?

$$I_e(\text{Marco}) = \text{Marco}$$

$$I_e(\text{Luigi}) = \text{Luigi}$$

$$I_e(\text{Alice}) = \text{Alice}$$

$$I_E(\text{Woman}) = \text{Woman}$$

$$I_E(\text{Student}) = \text{Student}$$

$$I_E(\text{Man}) = \text{Man}$$

$$I_o(\text{motherOf}) = \text{motherOf}$$

$$I_o(\text{fatherOf}) = \text{fatherOf}$$

$$\text{Man}(\text{Marco})$$

$$\text{Student}(\text{Luigi})$$

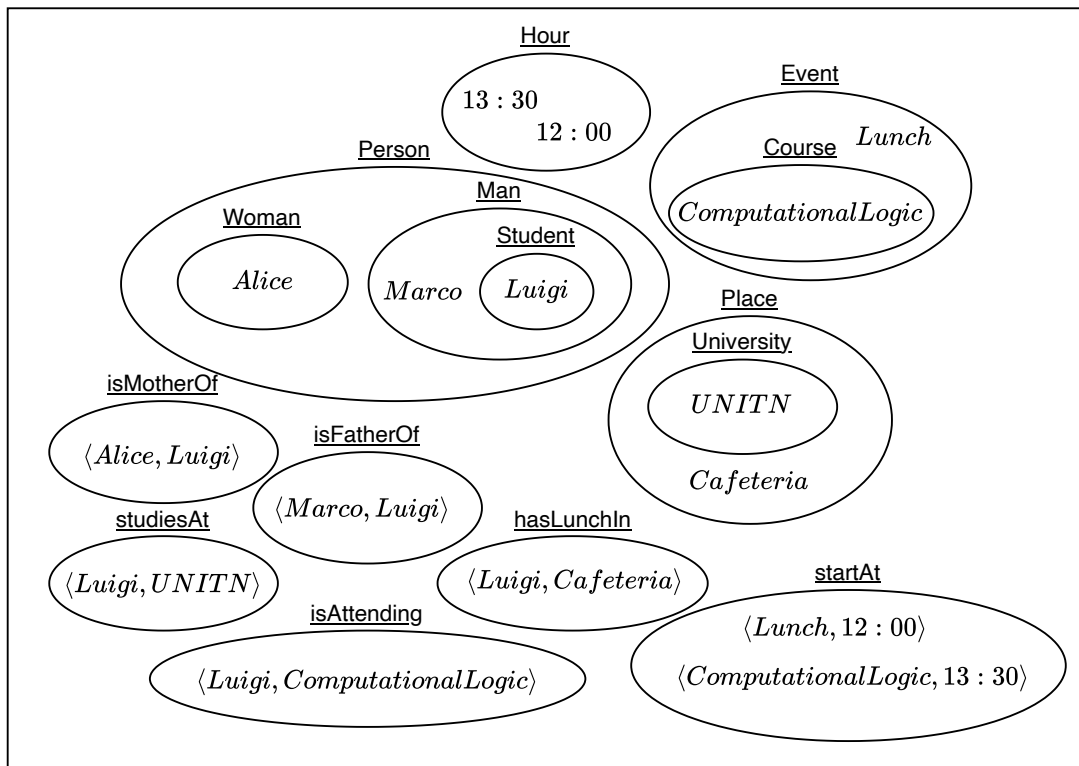
$$\text{Woman}(\text{Alice})$$

$$\text{motherOf}(\text{Alice}, \text{Luigi})$$

$$\text{faterOf}(\text{Marco}, \text{Luigi})$$

Yes, as every assertion of the theory is true in the model.

Ex2 - Correctness of a Theory



Considering the intended model on the left and the interpretation function below, is the following a theory correct?

$$I_e(\text{Marco}) = \text{Marco}$$

$$I_e(\text{Luigi}) = \text{Luigi}$$

$$I_e(\text{Alice}) = \text{Alice}$$

$$I_E(\text{Woman}) = \text{Woman}$$

$$I_E(\text{Student}) = \text{Student}$$

$$I_E(\text{Man}) = \text{Man}$$

$$I_o(\text{motherOf}) = \text{motherOf}$$

$$I_o(\text{fatherOf}) = \text{fatherOf}$$

$$\text{Student}(\text{Marco})$$

$$\text{Man}(\text{Luigi})$$

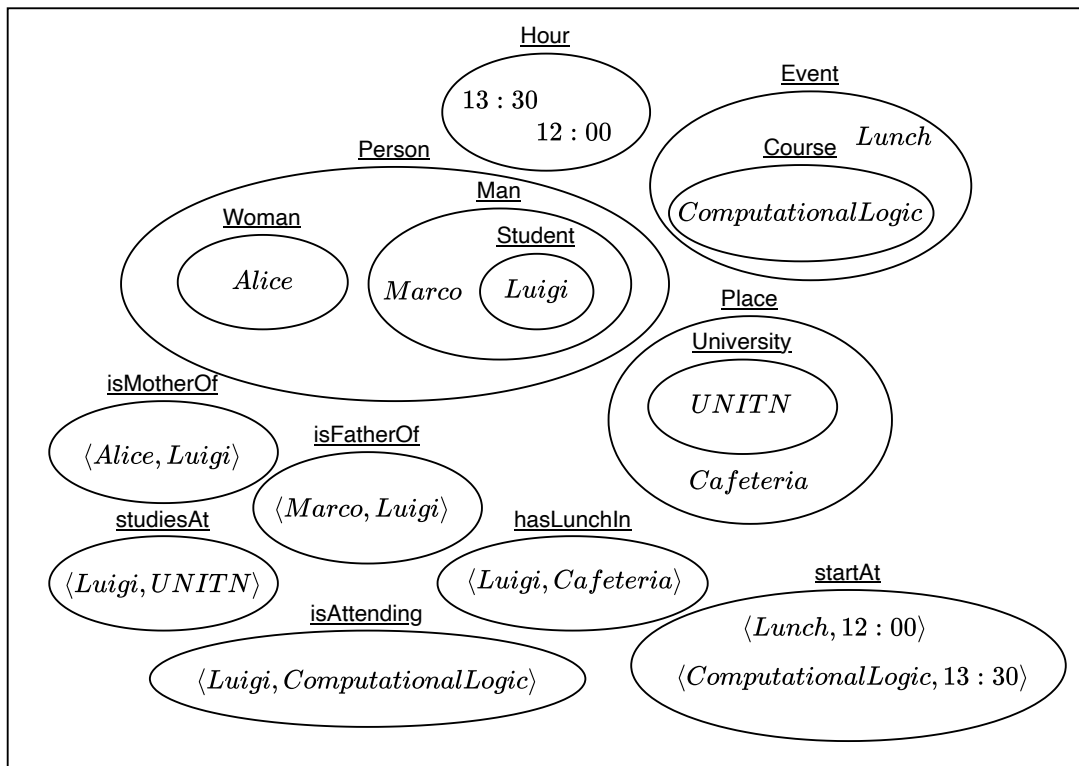
$$\text{Woman}(\text{Alice})$$

$$\text{motherOf}(\text{Alice}, \text{Luigi})$$

$$\text{faterOf}(\text{Marco}, \text{Luigi})$$

No, because Marco is not a student.

Ex3 - Correctness of a Theory



Considering the intended model on the left and the interpretation function below, is the following a theory correct?

$$I_e(\text{Marco}) = \text{Luigi}$$

$$I_e(\text{Luigi}) = \text{Marco}$$

$$I_e(\text{Alice}) = \text{Alice}$$

$$I_E(\text{Woman}) = \text{Woman}$$

$$I_E(\text{Student}) = \text{Student}$$

$$I_E(\text{Man}) = \text{Man}$$

$$I_o(\text{motherOf}) = \text{motherOf}$$

$$I_o(\text{fatherOf}) = \text{fatherOf}$$

$$\text{Student}(\text{Marco})$$

$$\text{Man}(\text{Luigi})$$

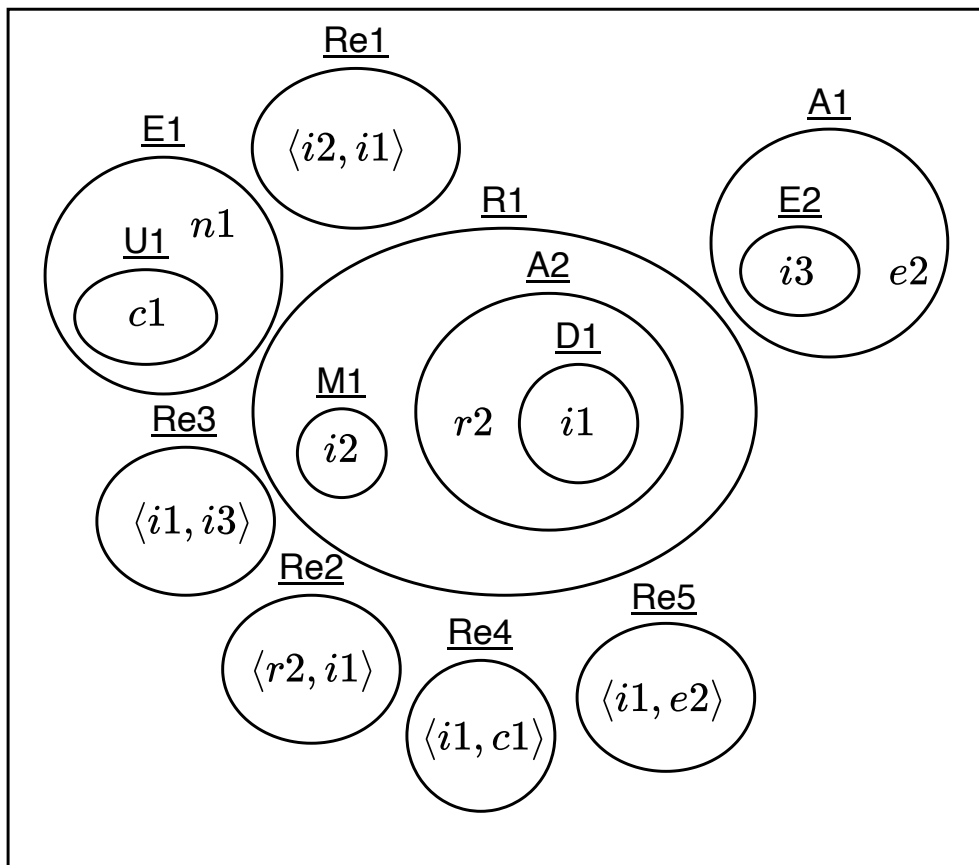
$$\text{Woman}(\text{Alice})$$

$$\text{motherOf}(\text{Alice}, \text{Marco})$$

$$\text{faterOf}(\text{Luigi}, \text{Marco})$$

Yes, as by the interpretation function «Marco» in the theory is «Luigi» in the model, and «Luigi» in the theory is «Marco» in the model.

Ex4 - Correctness of a Theory



Considering the intended model on the left and the interpretation function below, is the following a theory correct?

$$I_e(L) = i1$$

$$I_e(N) = i3$$

$$I_e(G) = c1$$

$$I_o(sA) = Re3$$

$$I_o(iA) = Re4$$

$$I_E(U) = E2$$

$$I_E(S) = D1$$

$$I_E(C) = U1$$

$$S(L)$$

$$sA(L, N)$$

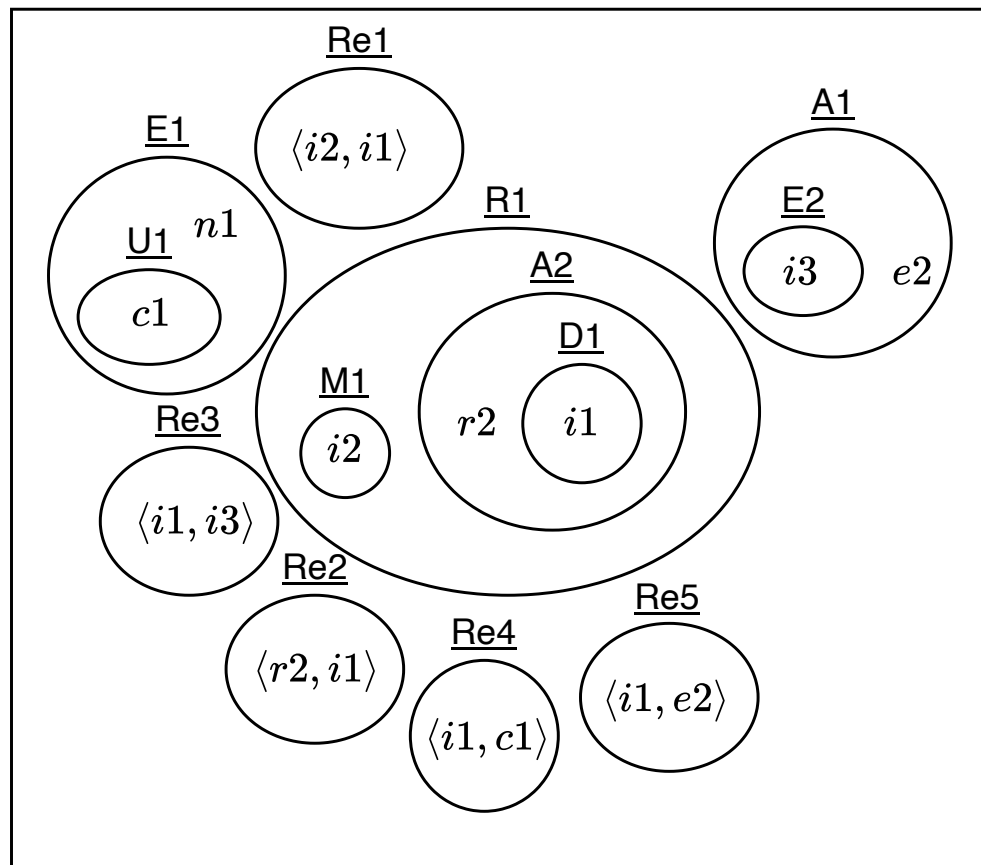
$$U(N)$$

$$iA(L, G)$$

$$C(G)$$

Yes, the model is actually the same as before, without the hours, and the theory has been abbreviated to single letters.

Ex5 - Correctness of a Theory



Considering the intended model on the left and the interpretation function below, is the following a theory correct?

$$I_e(L) = i1 \quad I_E(U) = E2$$

$$I_e(N) = c1 \quad I_E(S) = D1$$

$$I_e(G) = i3 \quad I_E(C) = U1$$

$$I_o(sA) = Re3$$

$$I_o(iA) = Re4$$

$$S(L)$$

$$sA(L, N)$$

$$U(N)$$

$$iA(L, G)$$

$$C(G)$$

No, as $\langle i1, c1 \rangle \notin Re3$ and $\langle i1, i3 \rangle \notin Re4$.

Ex 6 – Define the interpretation function

Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

No, because in E the integers must be represented as integers (integer), where name and value must coincide; this is true for all data values.

- $E = \text{MarioRossi\#1}, \text{StefaniaBianchi\#1}, \text{Trento\#1}, \text{Naples\#1}, \text{twenty-seven}, \text{thirty}$
- $\{C\} = \text{Student, entity, integer, dtype}$
- $\{R\} = \text{Domicile, Mark}$

Ex 7– Define the interpretation function

Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

- $E = \text{MarioRossi\#1}, \text{StefaniaBianchi\#1}, \text{Trento\#1}, \text{Naples\#1}, 27, 30$
- $\{C\} = \text{Student}, \text{city}, \text{entity}, \text{integer}, \text{dtype}$
- $\{R\} = \text{Domicile}, \text{Mark}$

Yes, because it is okay to add in D more elements than used in the table (a model is a subset of the domain), i.e. in the specific case “city” as etype intended even if not explicitly represented in the table; note that entity and dtype should always be put in C .

Ex 8– Define the interpretation function

Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

Yes, because cities here are represented as elements of the entity set, that is, the set containing all entities.

- $E = \text{MarioRossi, StefaniaBianchi, Trento, Naples, 27, 30}$
- $\{C\} = \text{Student, entity, integer, dtype}$
- $\{R\} = \text{Domicile, Mark}$

Ex 9 – Define the interpretation function

Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

Yes, because the domain can also have more elements than the table.

- $E = \text{MarioRossi\#1}, \text{StefaniaBianchi\#1}, \text{Trento\#1}, \text{Naples\#1}, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30$
- $\{C\} = \text{Student, entity, integer, dtype}$
- $\{R\} = \text{Domicile, Mark}$

Ex 10 – Define the interpretation function

Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

No, because an element for Naples is missing in E; a domain must have all the elements mentioned in the language, otherwise the corresponding interpretation cannot be defined.

- $E = \text{MarioRossi\#1}, \text{StefaniaBianchi\#1}, \text{Trento\#1}, 27, 30$
- $\{C\} = \text{Student}, \text{integer}, \text{entity}, \text{dtype}$
- $\{R\} = \text{Domicile}, \text{Mark}$

Ex 11 – Define the interpretation function

Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

Yes, because the interpretation function need not preserve names for entities, etypes or relations.

- $E = \text{MarioRossi\#1}, \text{StefaniaBianchi\#1}, \text{Bolzano\#1}, \text{Rome\#1}, 27, 30$
- $\{C\} = \text{Student}, \text{integer}, \text{entity}, \text{dtype}$
- $\{R\} = \text{Domicile}, \text{Mark}$

Ex 12 – Define the interpretation function

Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

No, because, given the table, domicile cannot be an etype.

- $E = MR, SB, TN, NA, 27, 30,$
- $\{C\} = \text{Student, Domicile, integer, entity, dtype}$
- $\{R\} = \text{Mark}$

Ex 13 – Define the interpretation function

Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

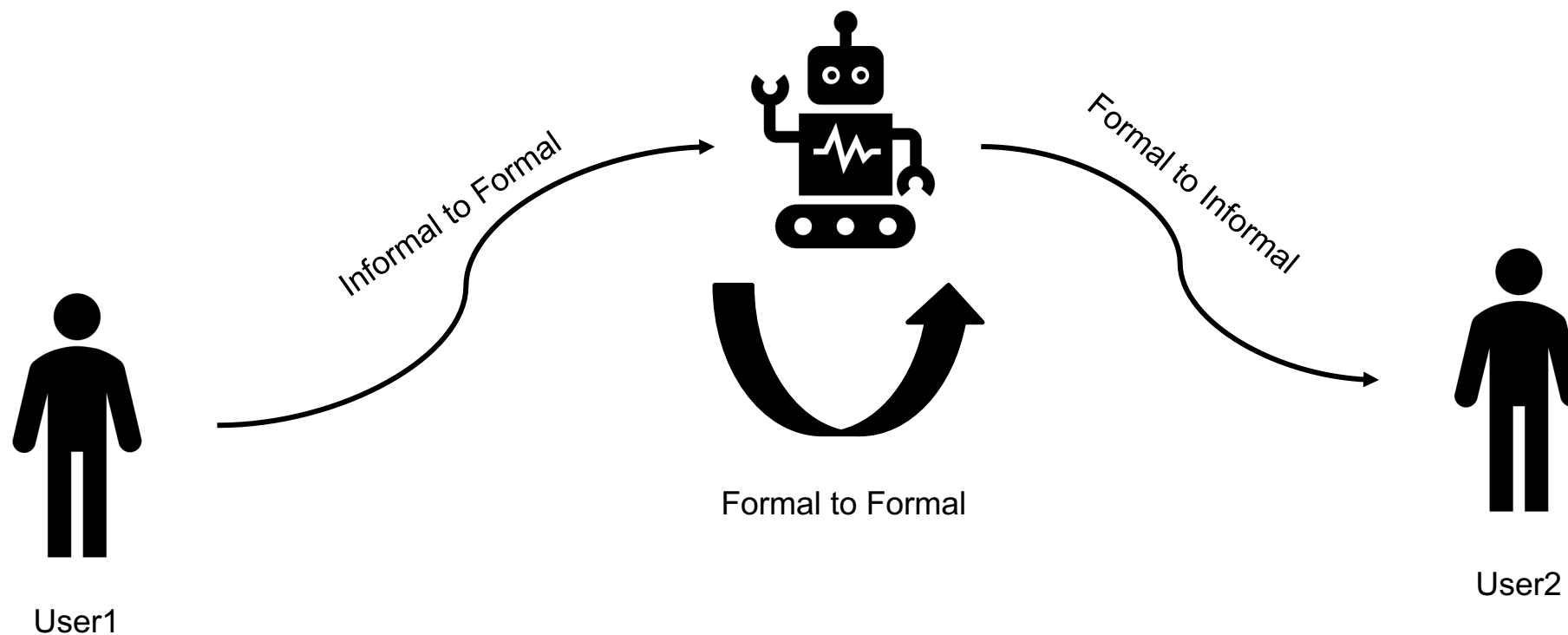
No, because the formalization of the domain is not complete; the R are missing.

- $E = \text{MarioRossi\#1}, \text{StefaniaBianchi\#1}, \text{Trento\#1}, \text{Naples\#1}, 27, 30,$
- $\{C\} = \text{Student}, \text{integer}, \text{entity}, \text{dtype}$

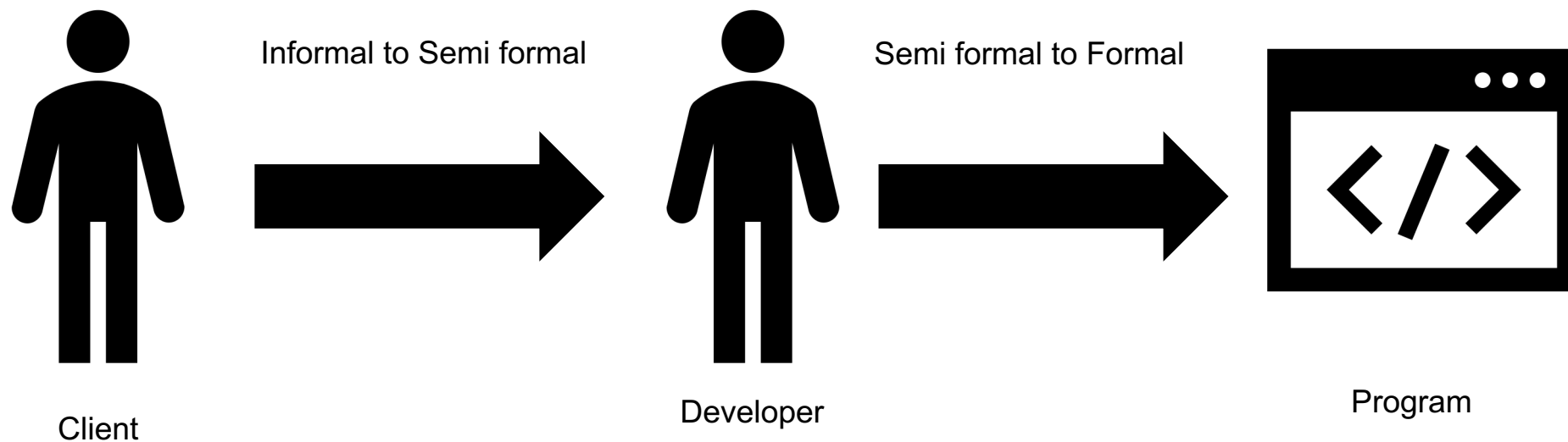
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Types of problems



Types of problems



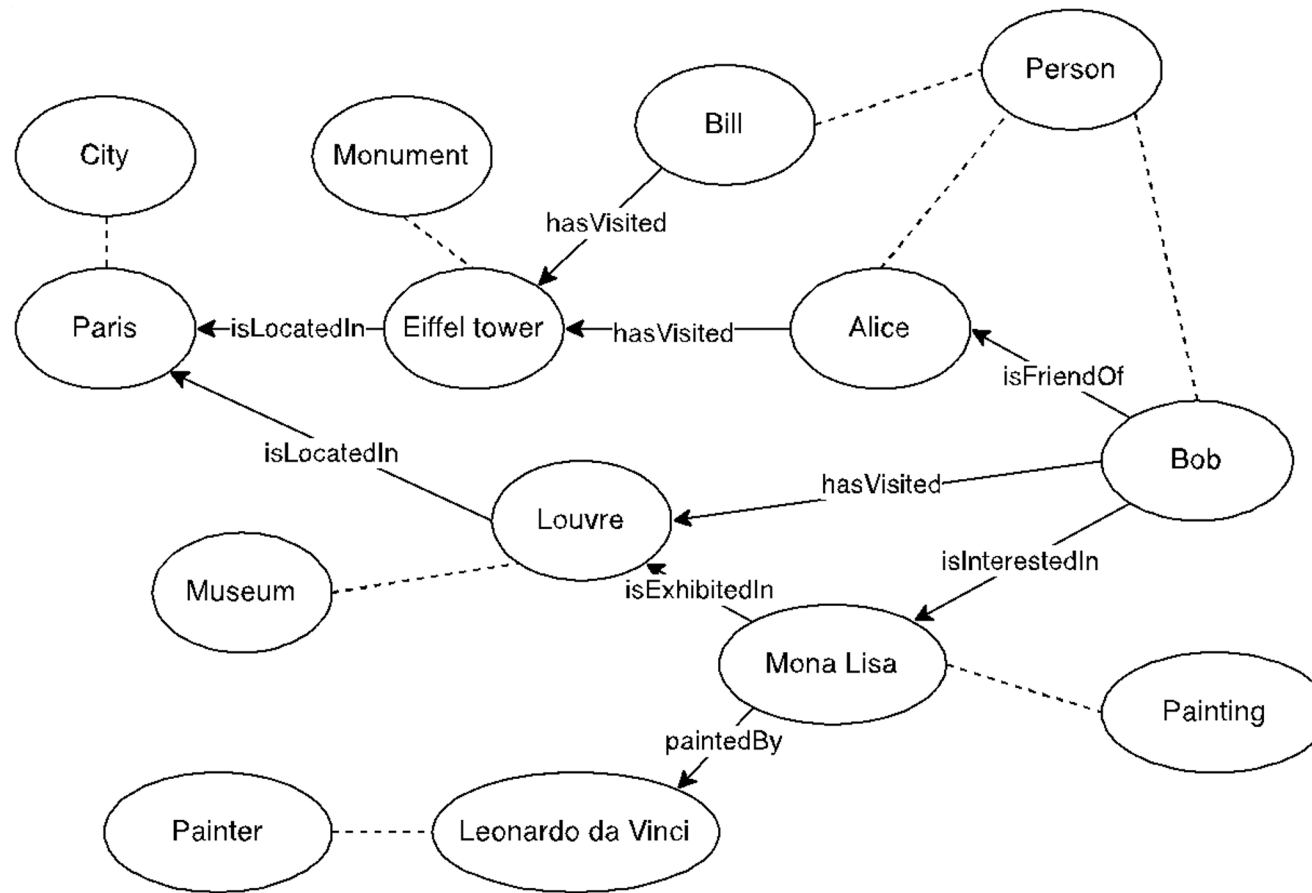
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Example 1 – From informal (NL) to semi-formal (KG)

Bill and Alice went to Paris with Alice's friend Bob. Bill and Alice visited the Eiffel Tower, one of the most visited monuments in Paris.

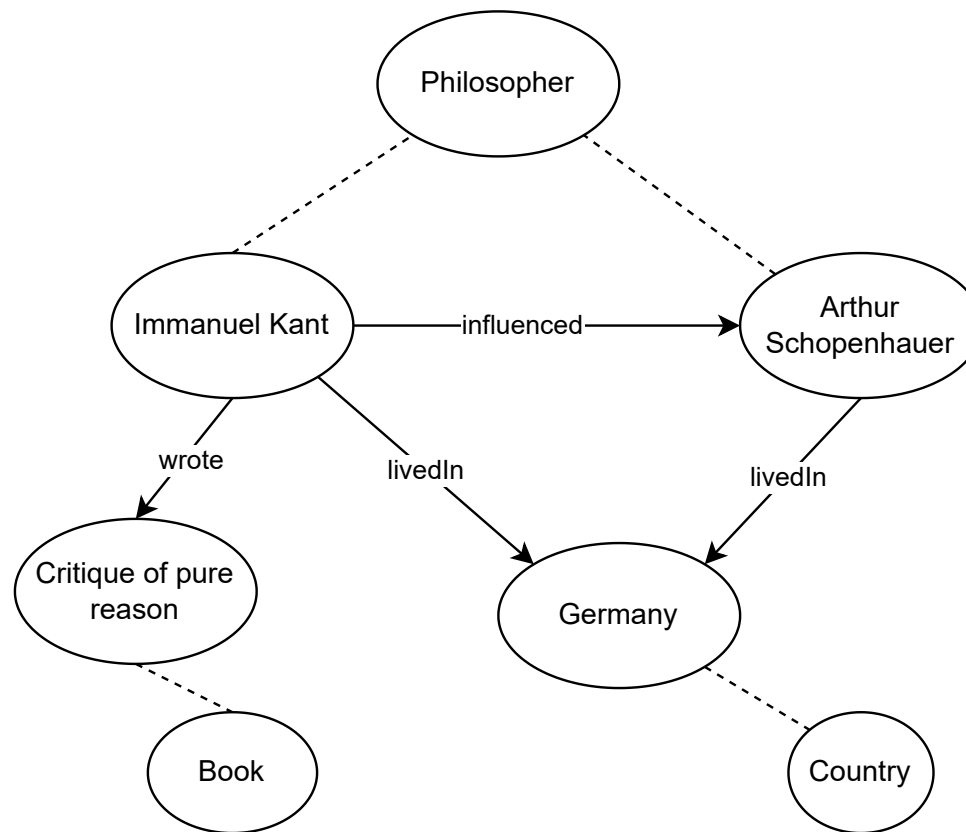
Instead, Bob was interested in the Mona Lisa, painted by painter Leonardo da Vinci, so he went to the Louvre, a famous museum in Paris.



Ex 14 - I2SF

Determine if the semi-formal representation on the right correctly represents the informal representation on the left.

"Immanuel Kant was a German philosopher best known for his work "Critique of Pure Reason". Arthur Schopenhauer was also a German philosopher who was influenced by Kant's ideas."



Yes, by using common sense, we can derive that Germany is a country, that Critique of Pure Reason is a book and that Kant wrote it.

Ex 15 - I2SF

Determine if the semi-formal representation on the right correctly represents the informal representation on the left.

”Immanuel Kant was a German philosopher best known for his work “Critique of Pure Reason”. Arthur Schopenhauer was also a German philosopher who was influenced by Kant’s ideas.”

Philosopher		
name	nationality	influenced by
Immanuel Kant	German	----
Arthur Schopenhauer	German	Immanuel Kant

Book	
title	author
Critique of Pure Reason	Immanuel Kant

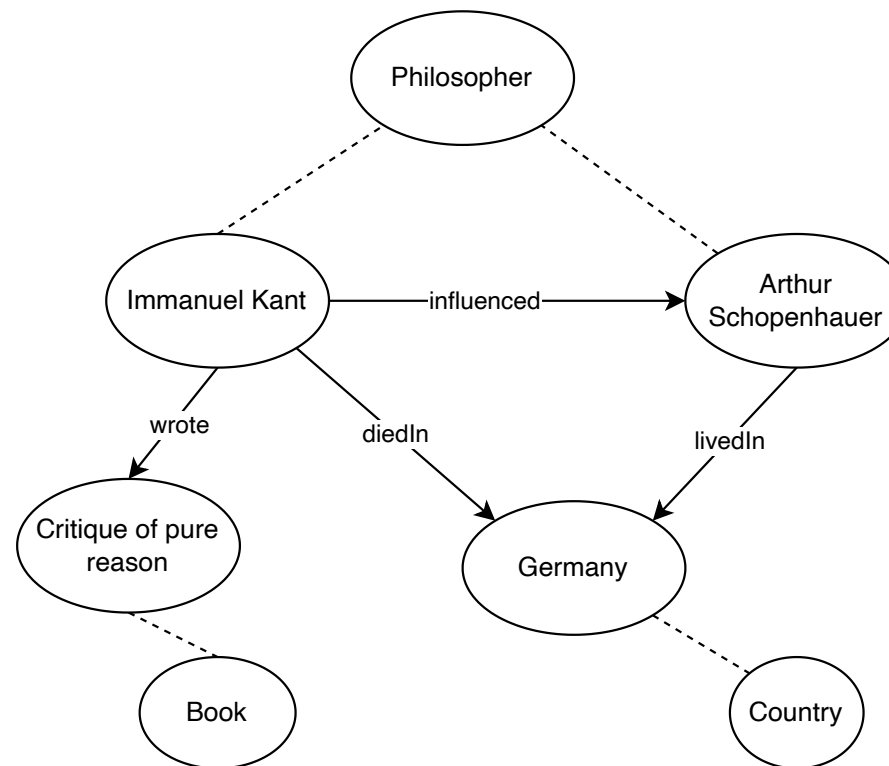
Yes, the representation on the right is essentially equivalent to the previous one.

Ex 16 - I2SF

Determine if the semi-formal representation on the right correctly represents the informal representation on the left.

"Immanuel Kant was a German philosopher best known for his work "Critique of Pure Reason". Arthur Schopenhauer was also a German philosopher who was influenced by Kant's ideas."

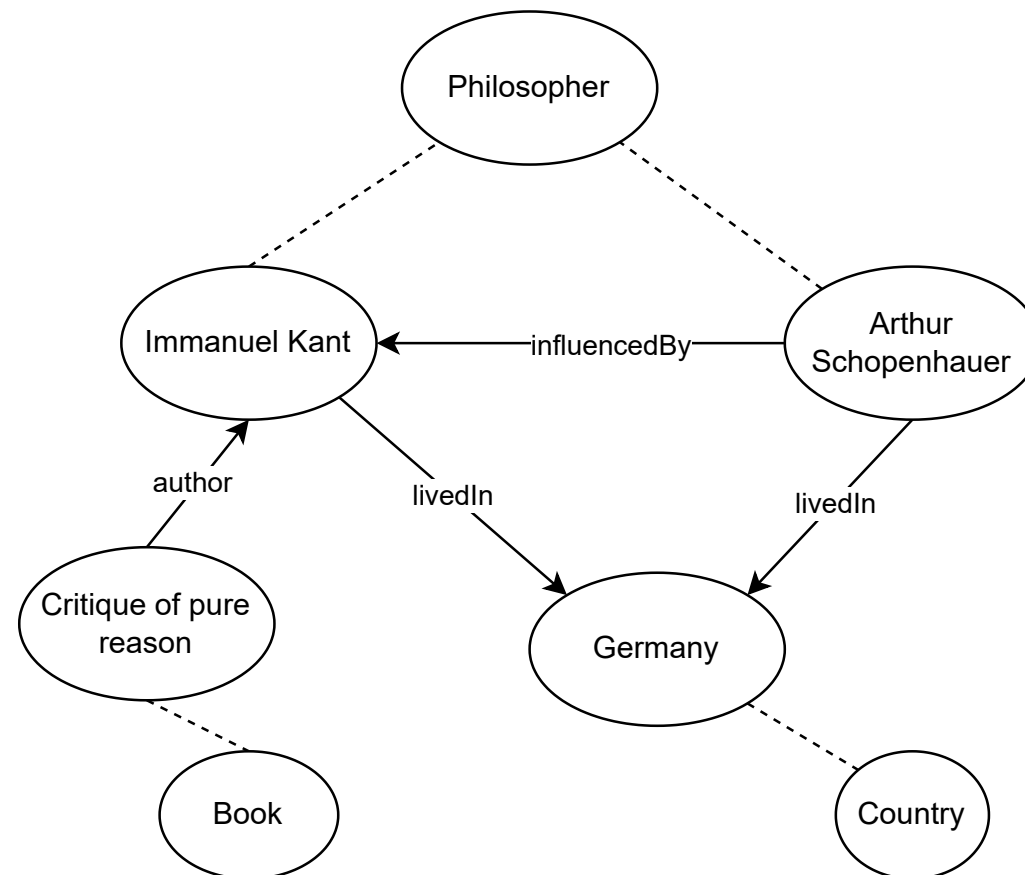
No, even if the informal representation says that Kant is German, it doesn't mean that we know that he died in Germany just from the given representation.



Ex 17 - I2SF

Determine if the semi-formal representation on the right correctly represents the informal representation on the left.

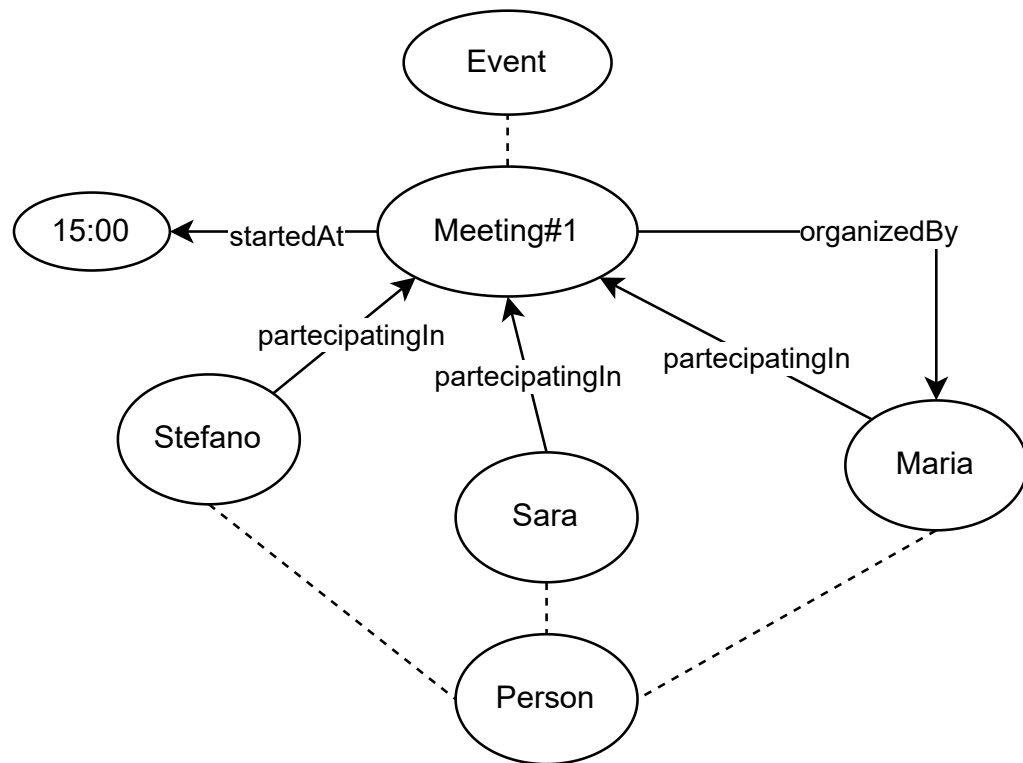
"Immanuel Kant was a German philosopher best known for his work "Critique of Pure Reason". Arthur Schopenhauer was also a German philosopher who was influenced by Kant's ideas."



Yes, even if the relationships are inverted, the labels used reflect the change.

Ex 18 - SF2I

Determine if the informal representation on the right correctly represents the semi-formal representation on the left.

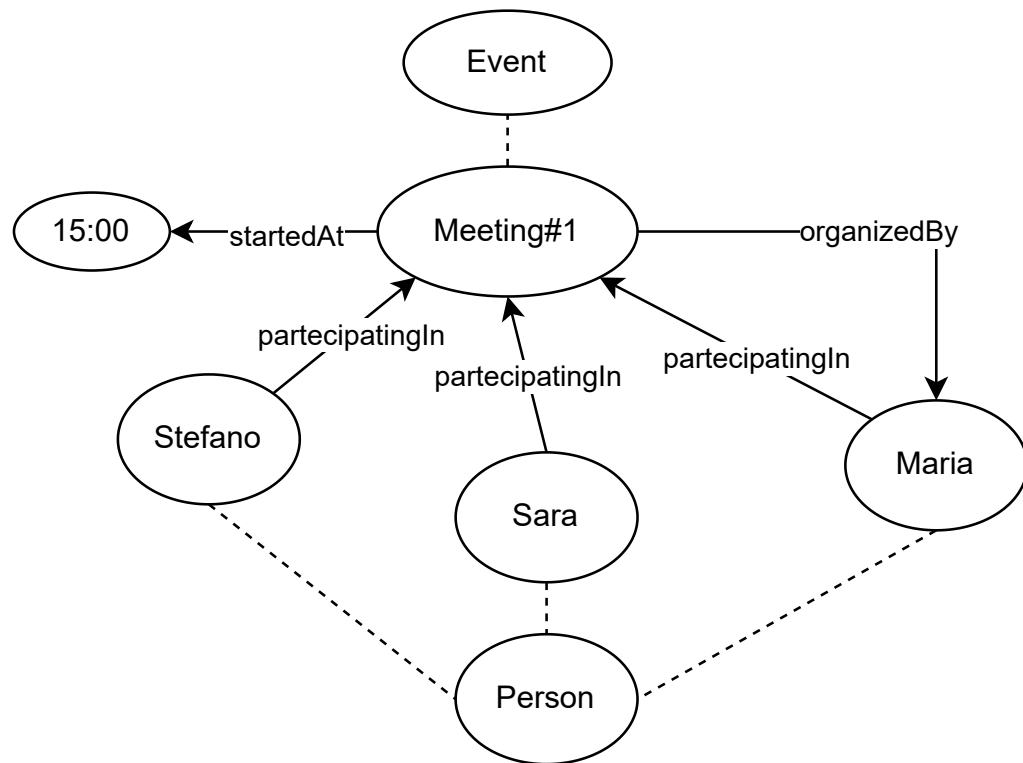


“Stefano, Maria, and Sara are participating in the meeting organized by Maria at 15:00.”

Yes, the informal representation describe completely the content of the graph.

Ex 19 - SF2I

Determine if the informal representation on the right correctly represents the semi-formal representation on the left.

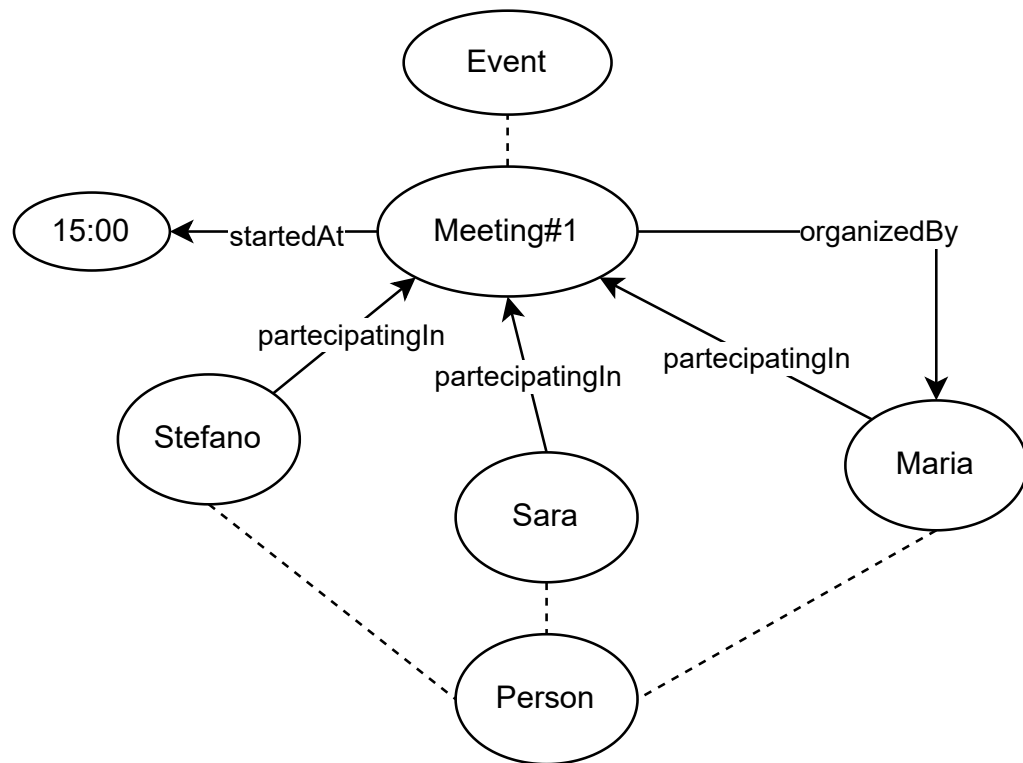


“Stefano and Sara are participating in the meeting organized by their boss, Maria.”

No, there is no reference to the fact that Maria is the boss of Stefano and Sara, and it's not derivable by common sense reasoning.

Ex 20 - SF2I

Determine if the informal representation on the right correctly represents the semi-formal representation on the left.



“Maria organized a meeting at 15:00.”

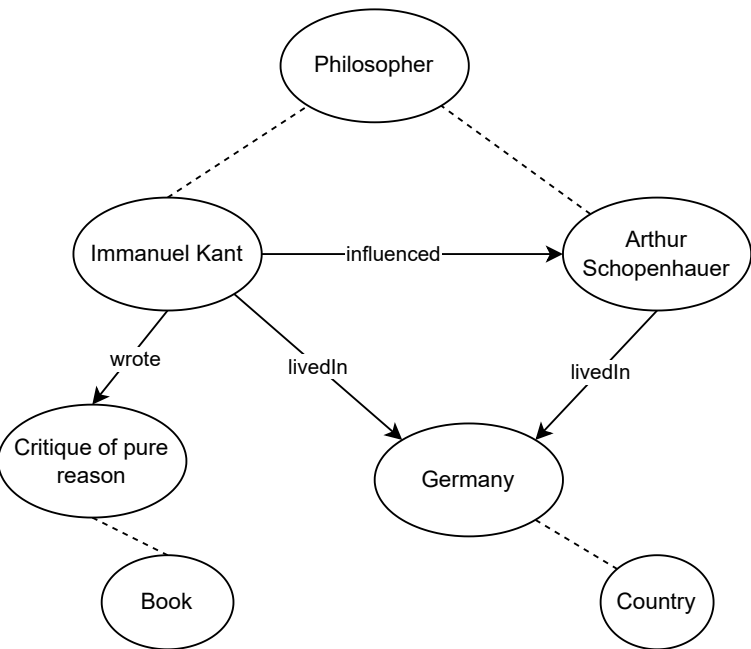
Yes, while the informal representation doesn't represent the graph completely, it is still correct.

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Ex 21 - SF2F

Determine if the formal representation on the right correctly represents the semi-formal representation on the left, considering the given interpretation function and model.



$I_e(Kant) = "ImmanuelKant"$

$I_e(Schopenhauer) = "ArthurSchopenhauer"$

$I_e(Germany) = "Germany"$

$I_e(Critique) = "CritiqueOfPureReason"$

$I_o(wrote) = "wrote"$

$I_o(influenced) = "influenced"$

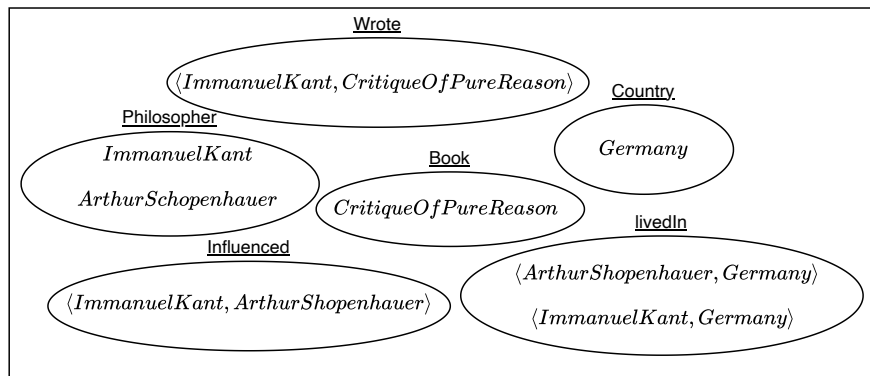
$I_o(livedIn) = "livedIn"$

$I_E(Book) = "Book"$

$I_E(Country) = "Country"$

$I_E(Philosopher) = "Philosopher"$

Yes, everything corresponds as expected.



$Philosopher(Kant)$

$Philosopher(Schopenhauer)$

$Country(Germany)$

$Book(Critique)$

$livedIn(Schopenhauer, Germany)$

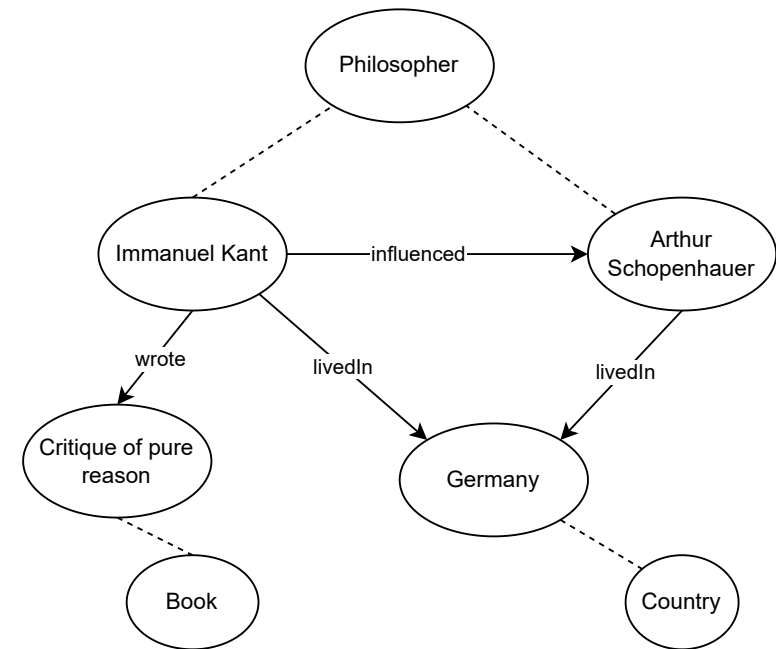
$livedIn(Kant, Germany)$

$influenced(Kant, Schopenhauer)$

$wrote(Kant, Critique)$

Ex 22 - SF2F

Determine if the formal representation on the right correctly represents the semi-formal representation on the left, considering the given interpretation function and model.



$I_e(Kant) = "ImmanuelKant"$

$I_e(Schopenhauer) = "ArthurSchopenhauer"$

$I_e(Germany) = "Germany"$

$I_e(Critique) = "CritiqueOfPureReason"$

$I_E(Book) = "Book"$

$I_E(Country) = "Country"$

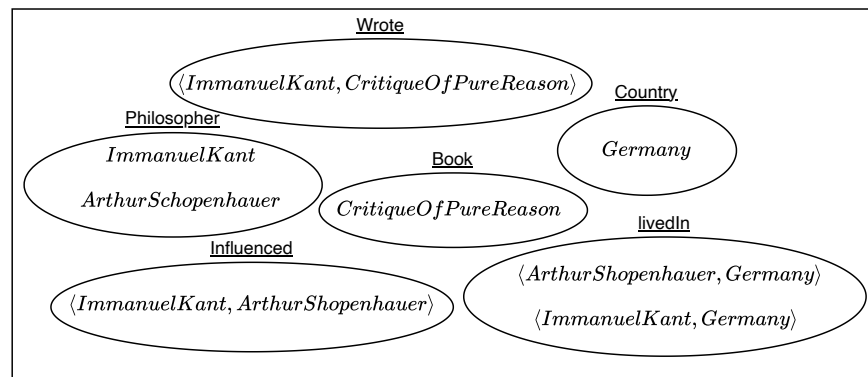
$I_E(Philosopher) = "Philosopher"$

$I_o(wrote) = "wrote"$

$I_o(influenced) = "influenced"$

$I_o(livedIn) = "livedIn"$

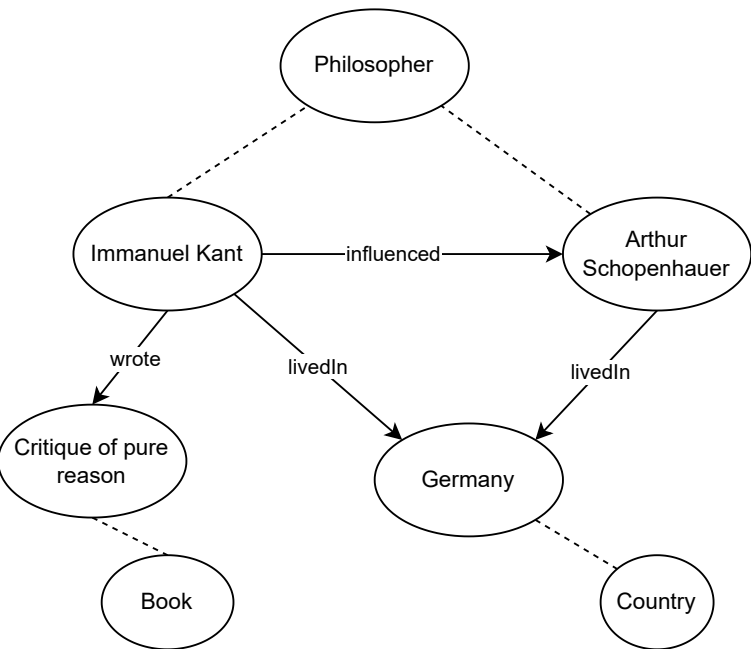
No, because there is no reference to the **authorOf** relation in the interpretation function.



$Philosopher(Kant)$
 $Philosopher(Schopenhauer)$
 $Country(Germany)$
 $Book(Critique)$
 $livedIn(Schopenhauer, Germany)$
 $livedIn(Kant, Germany)$
 $influenced(Kant, Schopenhauer)$
 $authorOf(Kant, Critique)$

Ex 23 - SF2F

Determine if the formal representation on the right correctly represents the semi-formal representation on the left, considering the given interpretation function and model.

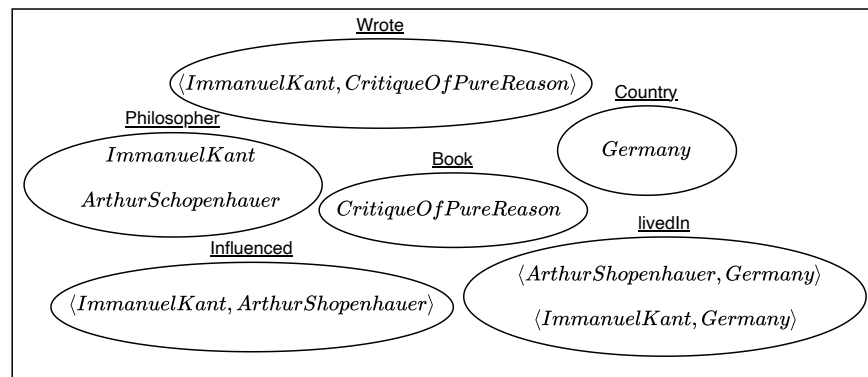


$I_e(Kant) = "ImmanuelKant"$
 $I_e(Schopenhauer) = "ArthurSchopenhauer"$
 $I_e(Germany) = "Germany"$
 $I_e(Critique) = "CritiqueOfPureReason"$

$I_E(Book) = "Book"$
 $I_E(Country) = "Country"$
 $I_E(Philosopher) = "Philosopher"$

$I_o(hasAuthor) = "wrote"$
 $I_o(influenced) = "influenced"$
 $I_o(livedIn) = "livedIn"$

Yes, even if it doesn't correspond to the semi formal representation, the interpretation function is connecting the has author relation to the wrote relation



$Philosopher(Kant)$
 $Philosopher(Schopenhauer)$
 $Country(Germany)$
 $Book(Critique)$
 $livedIn(Schopenhauer, Germany)$
 $livedIn(Kant, Germany)$
 $influenced(Kant, Schopenhauer)$
 $hasAuthor(Kant, Critique)$

Ex 24 - SF2F

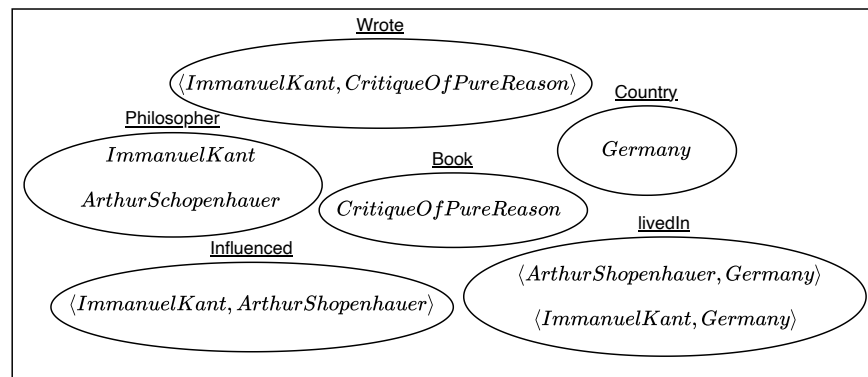
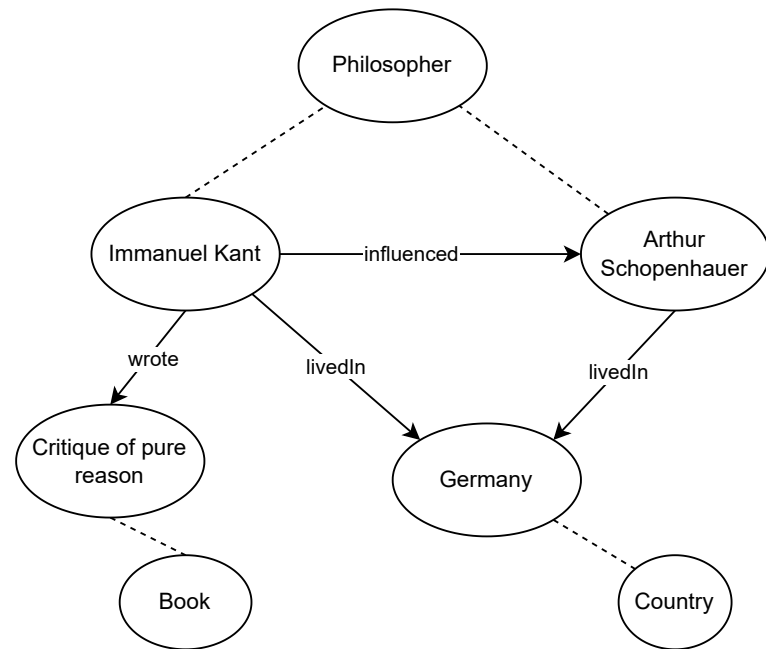
Determine if the formal representation on the right correctly represents the semi-formal representation on the left, considering the given interpretation function and model.

$I_e(K) = \text{"ImmanuelKant"}$
 $I_e(S) = \text{"ArthurSchopenhauer"}$
 $I_e(G) = \text{"Germany"}$
 $I_e(C) = \text{"CritiqueOfPureReason"}$

$I_o(hasAuthor) = \text{"wrote"}$
 $I_o(influenced) = \text{"influenced"}$
 $I_o(livedIn) = \text{"livedIn"}$

$I_E(Book) = \text{"Book"}$
 $I_E(Country) = \text{"Country"}$
 $I_E(Philosopher) = \text{"Philosopher"}$

No, hasAuthor relation is inverted

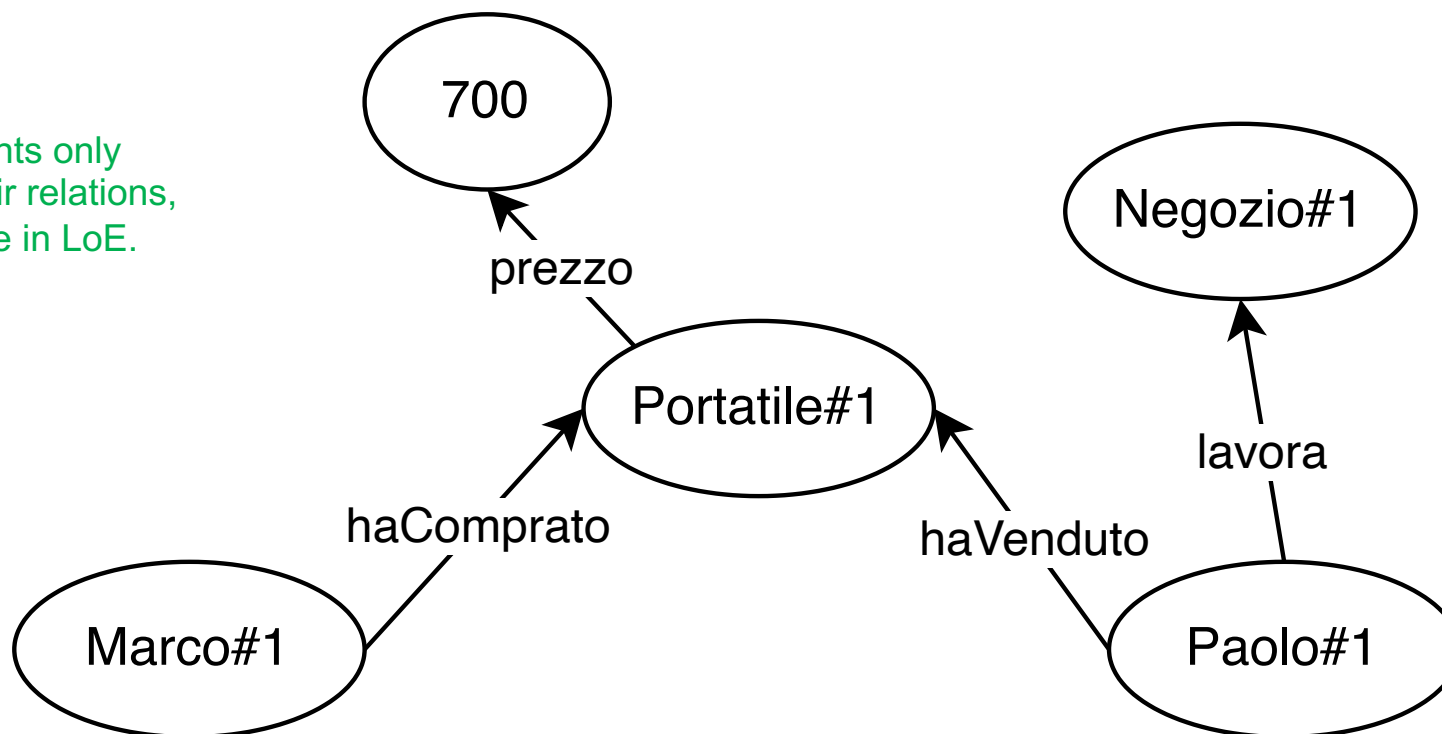


$Philosopher(K)$
 $Philosopher(S)$
 $Country(G)$
 $Book(C)$
 $livedIn(S, G)$
 $livedIn(K, G)$
 $influenced(K, S)$
 $hasAuthor(C, K)$

Ex 25 - SF2F

Determine if the informal knowledge graphs below can be fully formalized in LoE, without loss of information. Only the names of instances contain the symbol “#”; numbers must be considered as values of dtypes.

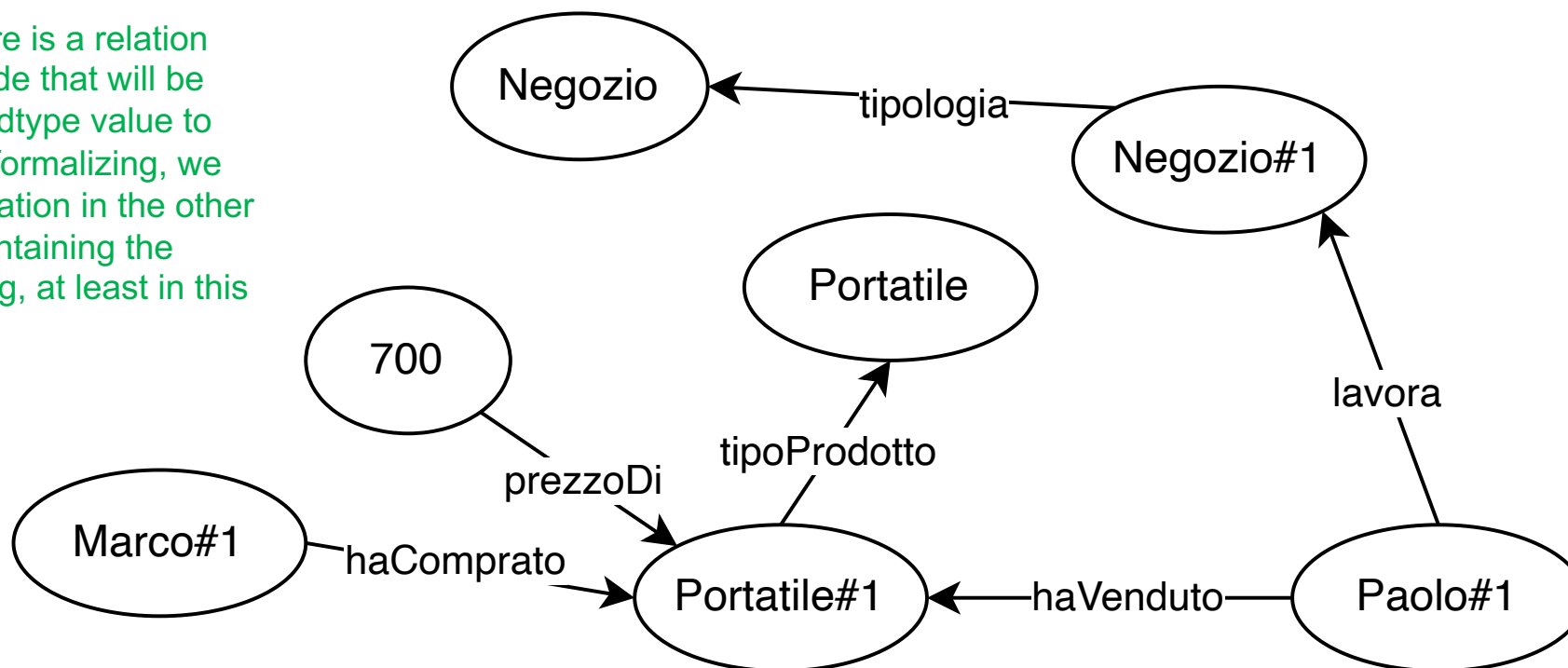
Yes, the graph represents only entities, values and their relations, therefore is formalizable in LoE.



Ex 26 - SF2F

Determine if the informal knowledge graphs below can be fully formalized in LoE, without loss of information. Only the names of instances contain the symbol “#”; numbers must be considered as values of dtypes.

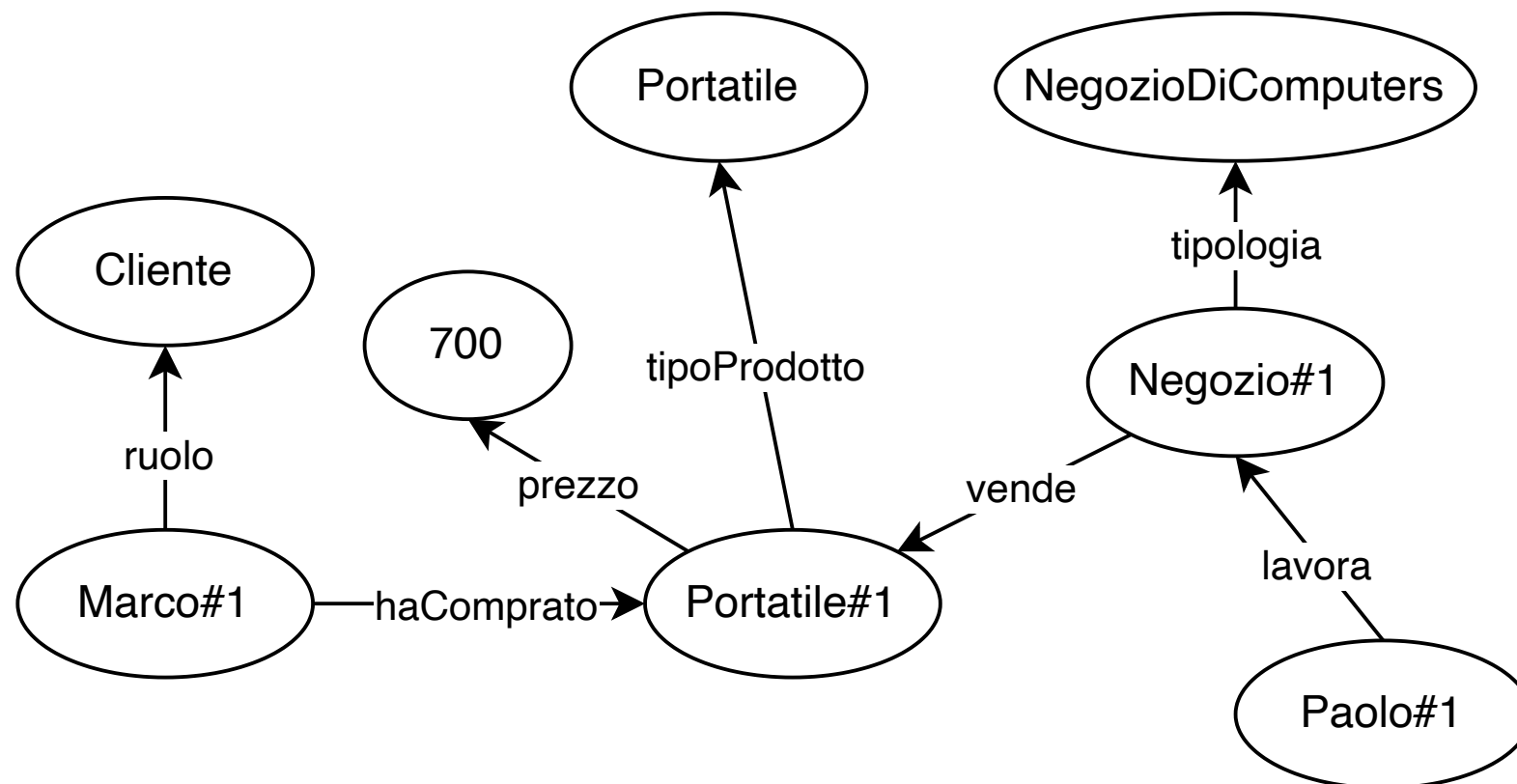
Yes, even if there is a relation going from a node that will be formalized as a dtype value to an entity, while formalizing, we can build the relation in the other direction by maintaining the informal meaning, at least in this case.



Ex 27 - SF2F

Determine if the informal knowledge graphs below can be fully formalized in LoE, without loss of information. Only the names of instances contain the symbol “#”; numbers must be considered as values of dtypes.

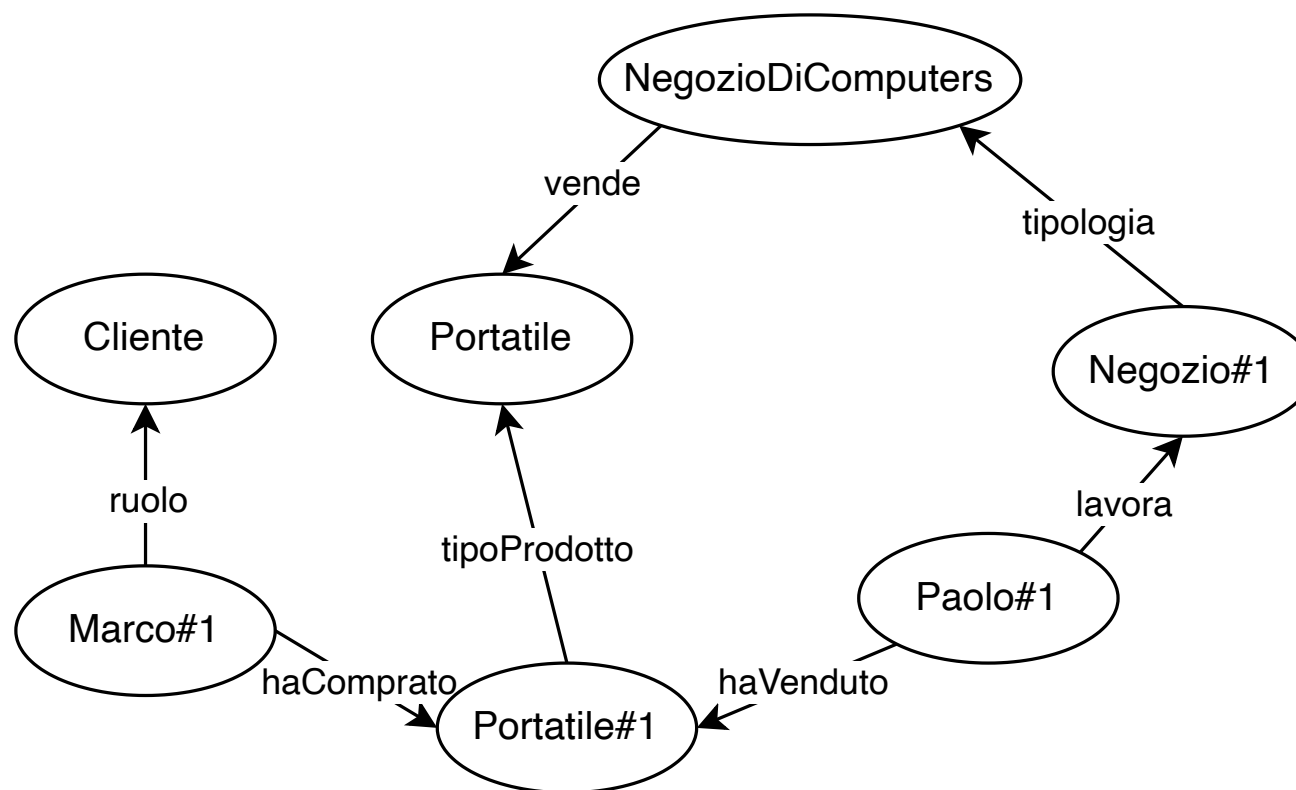
Yes, the graph is very similar to the first one, with the addition of etype information.



Ex 28 - SF2F

Determine if the informal knowledge graphs below can be fully formalized in LoE, without loss of information. Only the names of instances contain the symbol “#”; numbers must be considered as values of dtypes.

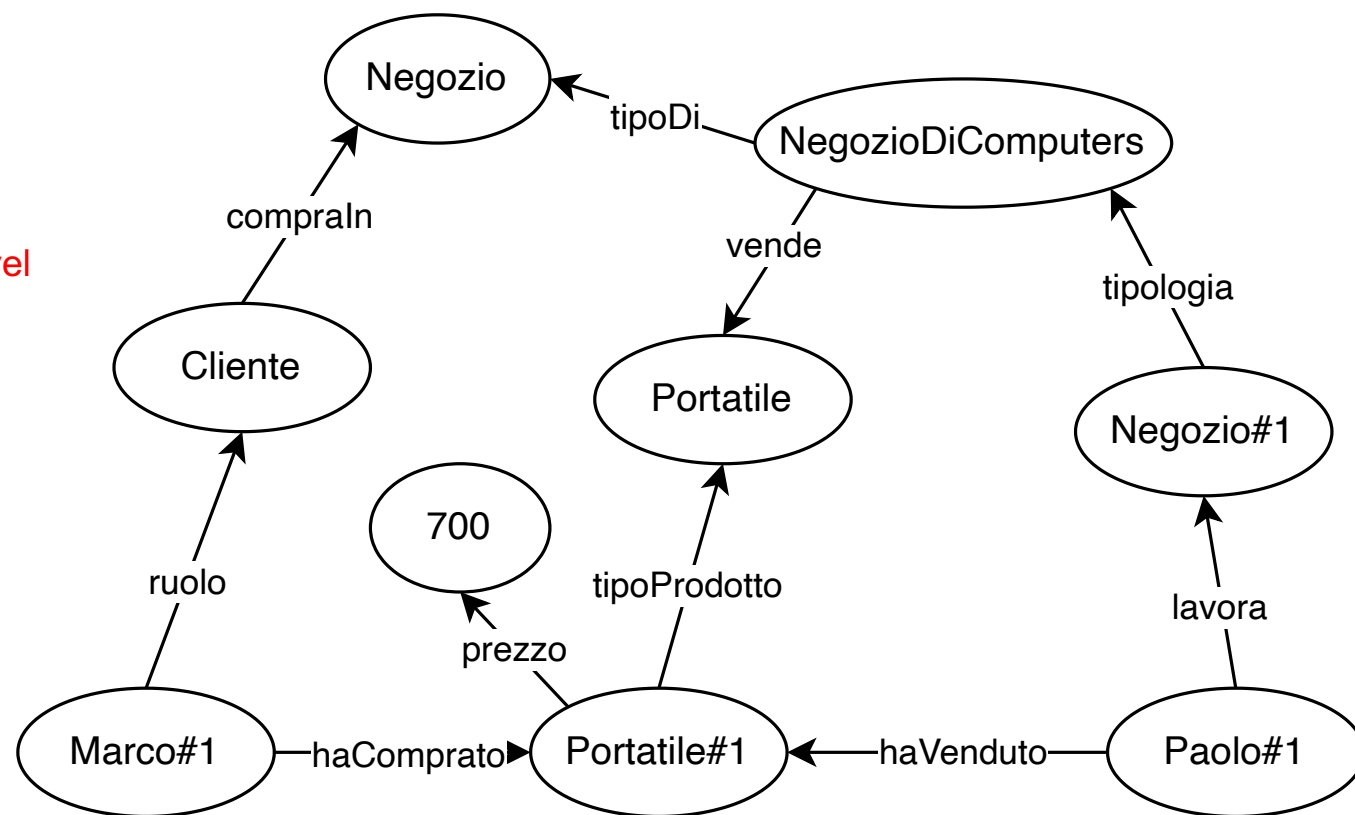
No, the graph represents knowledge level information, which cannot be formalized in LoE.



Ex 29 - SF2F

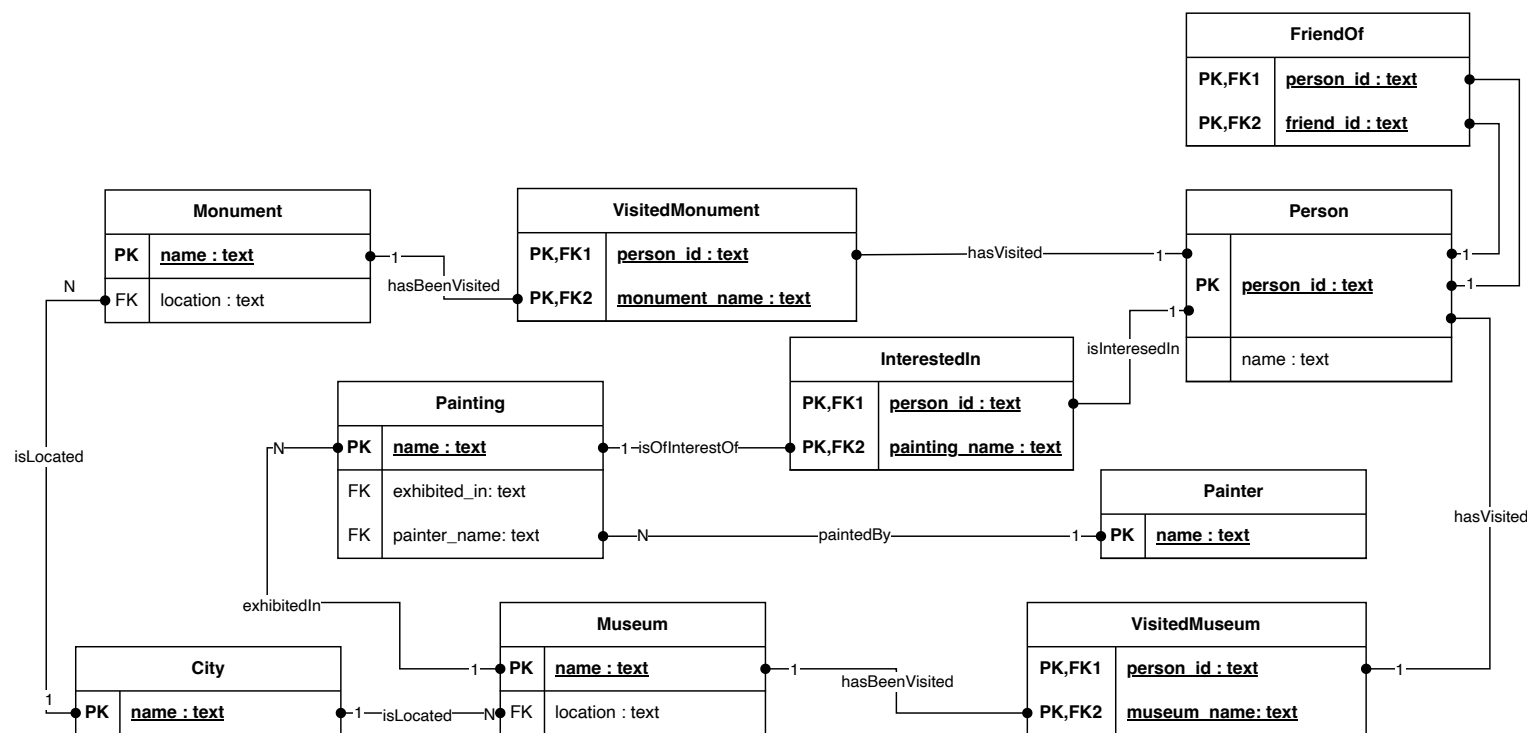
Determine if the informal knowledge graphs below can be fully formalized in LoE, without loss of information. Only the names of instances contain the symbol “#”; numbers must be considered as values of dtypes.

No, again, the graph represents knowledge level information.



Ex 30 - SF2F

Determine if the ER diagram below can be fully formalized in LoE, without loss of information.



No, an ER diagram represent only knowledge-level information, which can't be represented in LoE

Ex 31 - F2SF

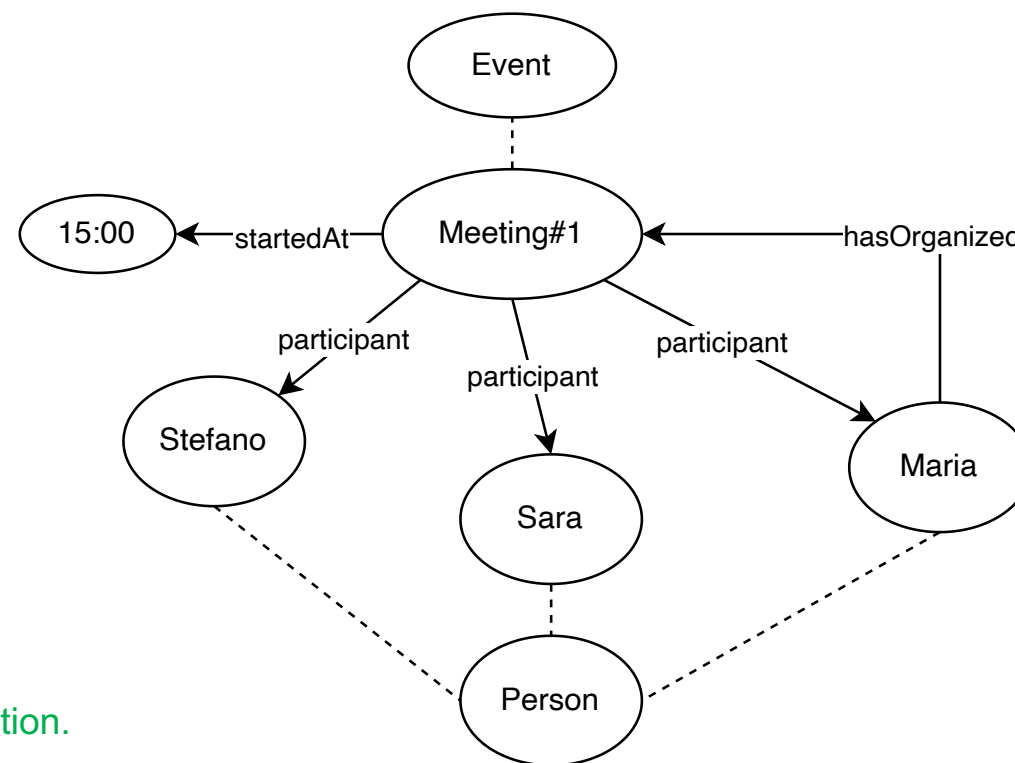
Determine if the semi-formal representation on the right is a correct translation of the formal representation on the left, considering the given interpretation function.

Person(Sara)
Person(Maria)
Person(Stefano)
Event(Meet#1)
partecipating(Sara, Meet#1)
partecipating(Stefano, Meet#1)
partecipating(Maria, Meet#1)
organizedBy(Meet#1, Maria)
startedAt(Meet#1, 15:00)
Hour(15:00)

$I_e(Sara) = "Sara"$
 $I_e(Stefano) = "Stefano"$
 $I_e(Maria) = "Maria"$
 $I_e(Meet\#1) = "Meeting\#1"$
 $I_v(15:00) = "15:00"$

$I_E(Event) = "Event"$
 $I_E(Person) = "Person"$
 $I_D(Hour) = "Hour"$

$I_o(partecipating) = "partecipatingIn"$
 $I_o(organizedBy) = "organizedBy"$
 $I_o(startedAt) = "startedAt"$



Yes, the meaning of the formal representation has been maintained in the translation.

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- **I2F and F2I**
- LoE entailment

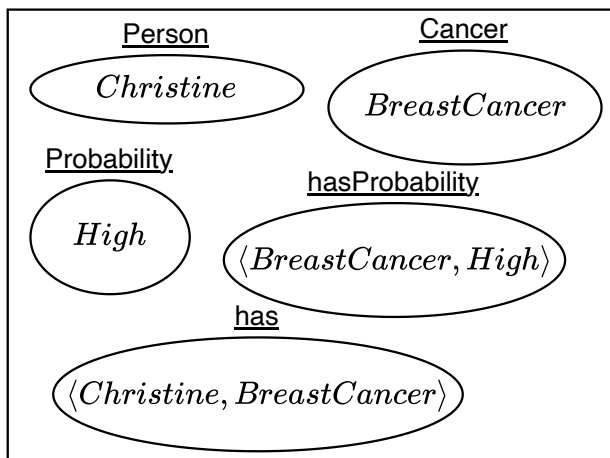
Ex 32 - I2F

Determine if the formalization in LoE of the informal representation on the left is correct.

“Christine has breast cancer with high probability”

No, while the formalized theory is consistent with the model and the interpretation function, it fails to represent correctly what the informal representation meant.

$I_e(\text{Christine}) = \langle \text{Christine} \rangle$
 $I_e(\text{BreastCancer}) = \langle \text{BreastCancer} \rangle$
 $I_v(\text{High}) = \langle \text{High} \rangle$
 $I_E(\text{Person}) = \langle \text{Person} \rangle$
 $I_E(\text{Cancer}) = \langle \text{Cancer} \rangle$
 $I_D(\text{Probability}) = \langle \text{Probability} \rangle$
 $I_O(\text{has}) = \langle \text{has} \rangle$
 $I_A(\text{hasProbability}) = \langle \text{hasProbability} \rangle$



$\text{Person}(\text{Christine})$
 $\text{Cancer}(\text{BreastCancer})$
 $\text{Probability}(\text{High})$
 $\text{has}(\text{Christine}, \text{BreastCancer})$
 $\text{hasProbability}(\text{BreastCancer}, \text{High})$

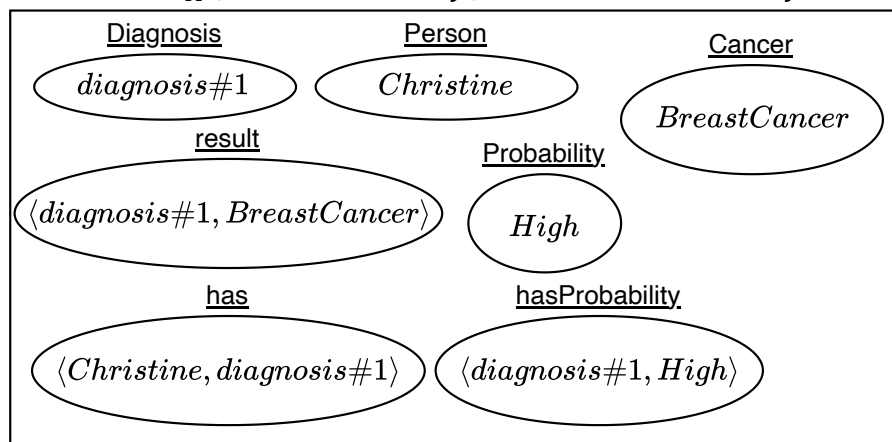
Ex 33 - I2F

Determine if the formalization in LoE of the informal representation on the left is correct.

“Christine has breast cancer with high probability”

Yes, in this case we have properly formalized the connection between Christine, the BreastCancer and its probability.

$I_e(Christine) = \langle Christine \rangle$
 $I_e(BreastCancer) = \langle BreastCancer \rangle$
 $I_e(diagnosis\#1) = \langle diagnosis\#1 \rangle$
 $I_v(High) = \langle High \rangle$
 $I_E(Person) = \langle Person \rangle$
 $I_E(Cancer) = \langle Cancer \rangle$
 $I_E(Diagnosis) = \langle Diagnosis \rangle$
 $I_D(Probability) = \langle Probability \rangle$
 $I_O(has) = \langle has \rangle$
 $I_O(result) = \langle result \rangle$
 $I_A(hasProbability) = \langle hasProbability \rangle$

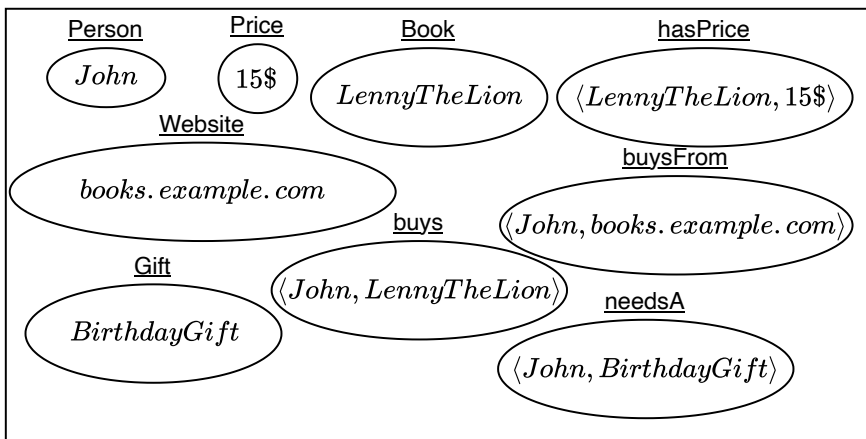


$Person(Christine)$
 $Cancer(BreastCancer)$
 $Probability(High)$
 $Diagnosis(diagnosis\#1)$
 $has(Christine, diagnosis\#1)$
 $result(diagnosis\#1, BreastCancer)$
 $hasProbability(diagnosis\#1, High)$

Ex 34 - I2F

Determine if the formalization in LoE of the informal representation on the left is correct.

"John buys a "Lenny the Lion" book from books.example.com for \$15 as a birthday gift"



$I_e(\text{John}) = \langle \text{John} \rangle$
 $I_e(\text{LennyTheLion}) = \langle \text{LennyTheLion} \rangle$
 $I_e(\text{booksExample}) = \langle \text{books.example.com} \rangle$
 $I_e(\text{BirthdayGift}) = \langle \text{BirthdayGift} \rangle$
 $I_v(15\$) = \langle 15\$ \rangle$
 $I_E(\text{Person}) = \langle \text{Person} \rangle$
 $I_E(\text{Book}) = \langle \text{Book} \rangle$
 $I_E(\text{Website}) = \langle \text{Website} \rangle$
 $I_E(\text{Gift}) = \langle \text{Gift} \rangle$
 $I_D(\text{Price}) = \langle \text{Price} \rangle$
 $I_O(\text{buys}) = \langle \text{buys} \rangle$
 $I_O(\text{buysFrom}) = \langle \text{buysFrom} \rangle$
 $I_O(\text{needsA}) = \langle \text{needsA} \rangle$
 $I_A(\text{hasPrice}) = \langle \text{hasPrice} \rangle$

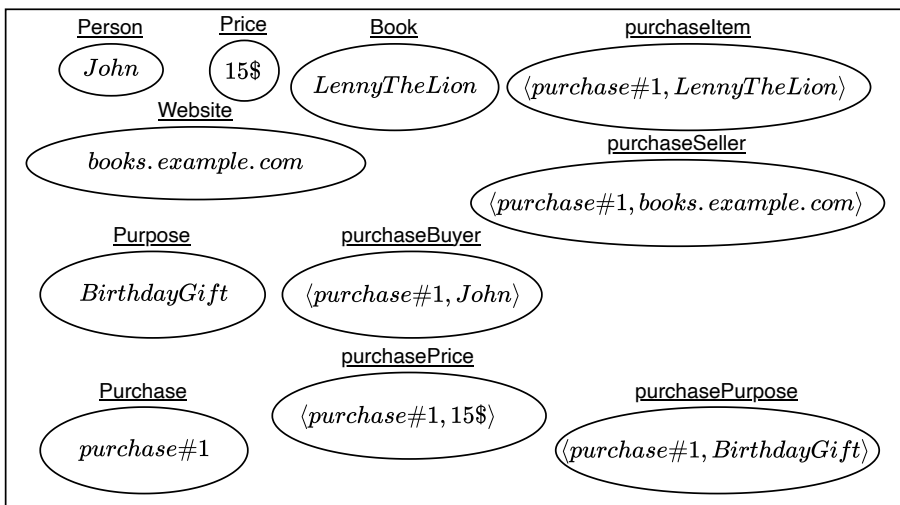
$\text{Person}(\text{John})$
 $\text{Book}(\text{LennyTheLion})$
 $\text{Website}(\text{booksExample})$
 $\text{Gift}(\text{BirthdayGift})$
 $\text{Price}(15\$)$
 $\text{hasPrice}(\text{LennyTheLion}, 15\$)$
 $\text{buys}(\text{John}, \text{LennyTheLion})$
 $\text{needsA}(\text{John}, \text{BirthdayGift})$
 $\text{buysFrom}(\text{John}, \text{booksExample})$

No, as before the formal representation fails to capture the meaning of the informal one. In fact it doesn't state John has bought the book as a gift, and that he has bought it for \$15 from the books.example.com website.

Ex 35 - I2F

Determine if the formalization in LoE of the informal representation on the left is correct.

"John buys a "Lenny the Lion" book from books.example.com for \$15 as a birthday gift"



$I_e(\text{John}) = \langle \text{John} \rangle$
 $I_e(\text{LennyTheLion}) = \langle \text{LennyTheLion} \rangle$
 $I_e(\text{booksExample}) = \langle \text{books.example.com} \rangle$
 $I_e(\text{BirthdayGift}) = \langle \text{BirthdayGift} \rangle$
 $I_e(\text{purchase\#1}) = \langle \text{purchase\#1} \rangle$
 $I_v(15\$) = \langle 15\$ \rangle$
 $I_E(\text{Person}) = \langle \text{Person} \rangle$
 $I_E(\text{Book}) = \langle \text{Book} \rangle$
 $I_E(\text{Website}) = \langle \text{Website} \rangle$
 $I_E(\text{Purpose}) = \langle \text{Purpose} \rangle$
 $I_E(\text{Purchase}) = \langle \text{Purchase} \rangle$
 $I_D(\text{Price}) = \langle \text{Price} \rangle$
 $I_o(\text{buyer}) = \langle \text{purchaseBuyer} \rangle$
 $I_o(\text{seller}) = \langle \text{purchaseSeller} \rangle$
 $I_o(\text{purpose}) = \langle \text{purchasePurpose} \rangle$
 $I_o(\text{item}) = \langle \text{purchaseItem} \rangle$
 $I_A(\text{price}) = \langle \text{purchasePrice} \rangle$

Person(John)
 Book(LennyTheLion)
 Website(booksExample)
 Purpose(BirthdayGift)
 Price(15\$)
 Purchase(purchase#1)
 price(purchase#1,15\$)
 item(purchase#1,LennyTheLion)
 purpose(purchase#1,BirthdayGift)
 seller(purchase#1,booksExample)
 buyer(purchase#1,John)

Yes, in this representation, we fixed the issues of the previous one.

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LoE Entailment

Proposition (LoE entailment). Let T_a be a LoE theory for a model M , with $T_a \subseteq L_a$. Let $\underline{a} \in L_a$ and $T_{\underline{a}} \subseteq L_a$ be respectively an **assertion question** and a **theory question**. Then:

$M \models T_{\underline{a}}$ if and only if $T_a \models T_{\underline{a}}$ if, for all $a \in T_{\underline{a}}$, $a \in T_a$

$M \models \{\underline{a}\}$ if and only if $T_a \models \{\underline{a}\}$, written $T_a \models \underline{a}$, if $\underline{a} \in T_a$

Proposition evidence (LoE entailment). The LoE world model is canonical. There is a one-to-one mapping between a LOE theory and a LoE model.

Observation (LoE entailment). The LoE world entailment can be seen as translating two questions: (1) the default one, whether $\underline{a}$ is true in the reference (world) model and (2), whether $\underline{a}$ was already known and as such, it was already in memory.

Observation (Truth, Falsity and memory). Based on the previous observation, saying that something is True amounts to saying that your memory says so. No real reasoning involved.

LoE Instance checking

Definition (LoE Instance checking) LoE instance checking checks whether an assertion is entailed by a LoE theory. We have the following four possible cases

$T \models E(e)$	returns yes	if $E(e) \in T_{\text{LoE}}$,	IDK otherwise
$T \models D(v)$	returns yes	if $D(v) \in T_{\text{LoE}}$,	IDK otherwise
$T \models O(e_1, e_2)$	returns yes	if $O(e_1, e_2) \in T_{\text{LoE}}$,	IDK otherwise
$T \models A(e, v)$	returns yes	if $A(e, v) \in T_{\text{LoE}}$,	IDK otherwise

Observation 1 (LoE Instance checking). It amounts to checking whether the input assertion, modulo synonyms, is in the theory which generated the model.

LoE Instance checking – closed world

Observation (Instance checking). Consider the following two variations of instance checking

$T \models E(e)$ returns “yes, $E(e)$ is true” if $E(e) \in T_{LoE}$
returns “IDK” otherwise

$T \models E(e)$ returns “yes, $E(e)$ is true” if $E(e) \in T_{LoE}$
returns “no, $E(e)$ is false” otherwise

Open World Assumption (OWA): The first version, LoE instance checking, returns «I don’t know» when entailment does not hold.

Closed World Assumption /CWA): The second version, returns that « $E(e)$ is false» when entailment does not hold.

Observation. CWA transforms **partial knowledge** (a fact not being in the model) into **negative knowledge**. What is not known is assumed to be false. OWA is mainly applied in the Web, CWA is mainly applied corporate DBs.

LoE – Instance retrieval

Definition (LoE Instance retrieval). Given an etype (or object/ data property), LoE Instance retrieval returns all the entities (or pair “entity, entity/data”) in the memory T which satisfy the etype (or object property/attribute) in input

$$T \models E$$

$$T \models D$$

$$T \models O$$

$$T \models A$$

Observation (Instance retrieval). It generates all the possible assertions using the entities and values of the alphabet and applies model checking to the resulting assertion. It returns the assertions for which model checking returns “yes”.

Observation (Instance retrieval). Instance retrieval consists of a multiple call to Instance checking. Similar to Q/A in relational DBs.

Ex 36 - Instance Checking

Person(Alice)

Person(Bob)

Person(Bill)

City(Paris)

Monument(EiffelTower)

Museum(Louvre)

Painting(MonaLisa)

Painter(LeonardoDaVinci)

isFriendOf(Bob, Alice)

hasVisited(Bill, EiffelTower)

hasVisited(Bob, Louvre)

hasTraveledTo(Bob, Paris)

paintedBy(MonaLisa, LeonardoDaVinci)

exhibitedIn(MonaLisa, Louvre)

interestedIn(Bob, MonaLisa)

isLocatedIn(EiffelTower, Paris)

isLocatedIn(Louvre, Paris)

Considering the LoE theory T on the left, determine the results of the following instance checking statements. Assume OWA (open world assumption)

$T \models \text{paintedBy}(\text{MonaLisa}, \text{LeonardoDaVinci})$ True

$T \models \text{Person}(\text{Bob})$ True

$T \models \text{hasVisited}(\text{Bill}, \text{EiffelTower})$ True

$T \models \text{hasTraveledTo}(\text{Bill}, \text{Paris})$ Idk

$T \models \text{hasTraveledTo}(\text{Bob}, \text{Paris})$ True

$T \models \text{Person}(\text{LeonardoDaVinci})$ Idk

Ex 37 - Instance Checking

Person(Alice)

Person(Bob)

Person(Bill)

City(Paris)

Monument(EiffelTower)

Museum(Louvre)

Painting(MonaLisa)

Painter(LeonardoDaVinci)

isFriendOf(Bob, Alice)

hasVisited(Bill, EiffelTower)

hasVisited(Bob, Louvre)

hasTraveledTo(Bob, Paris)

paintedBy(MonaLisa, LeonardoDaVinci)

exhibitedIn(MonaLisa, Louvre)

interestedIn(Bob, MonaLisa)

isLocatedIn(EiffelTower, Paris)

isLocatedIn(Louvre, Paris)

Considering the LoE theory T on the left, determine the results of the following instance checking statements. Assume CWA (close world assumption)

$T \models isLocatedIn(Louvre, Paris)$ True

$T \models exhibitedIn(MonaLisa, Louvre)$ True

$T \models isLocatedIn(MonaLisa, Paris)$ False

$T \models isFriendOf(Alice, Bill)$ False

$T \models hasVisited(Bob, Louvre)$ True

$T \models traveledTogether(Bob, Bill)$ False

Ex 37 - Instance Retrieval

Person(Alice)

Person(Bob)

Person(Bill)

City(Paris)

Monument(EiffelTower)

Museum(Louvre)

Painting(MonaLisa)

Painter(LeonardoDaVinci)

isFriendOf(Bob, Alice)

hasVisited(Bill, EiffelTower)

hasVisited(Bob, Louvre)

hasTraveledTo(Bob, Paris)

paintedBy(MonaLisa, LeonardoDaVinci)

exhibitedIn(MonaLisa, Louvre)

interestedIn(Bob, MonaLisa)

isLocatedIn(EiffelTower, Paris)

isLocatedIn(Louvre, Paris)

Considering the LoE theory T on the left, determine the results of the following instance retrieval statements.

$T \models \textit{Person} \quad \{Alice, Bob, Bill\}$

$T \models \textit{City} \quad \{Paris\}$

$T \models \textit{Location} \quad \{\}$

$T \models \textit{hasVisited} \quad \{\langle Bill, EiffelTower \rangle, \langle Bob, Louvre \rangle\}$



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LoE (T2MP)